

MCGILL INTERNATIONAL PORTFOLIO CHALLENGE 2023

Maple Leaf Pension Plan

*Developing an Inflationary Resilient
Decumulation Strategy*



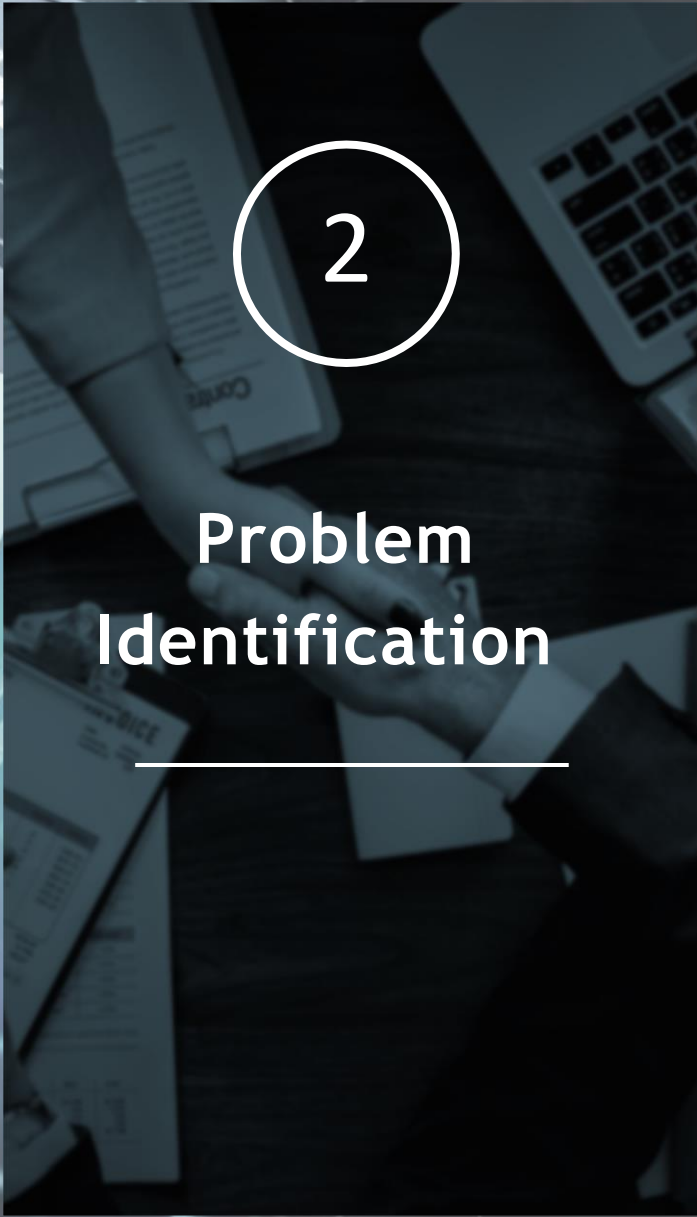
KIPS CONSULTING

Investment Consultancy working with MLPP

A dark, stylized map of Europe with various countries and cities labeled. The map is overlaid with a white circle containing the number 1.

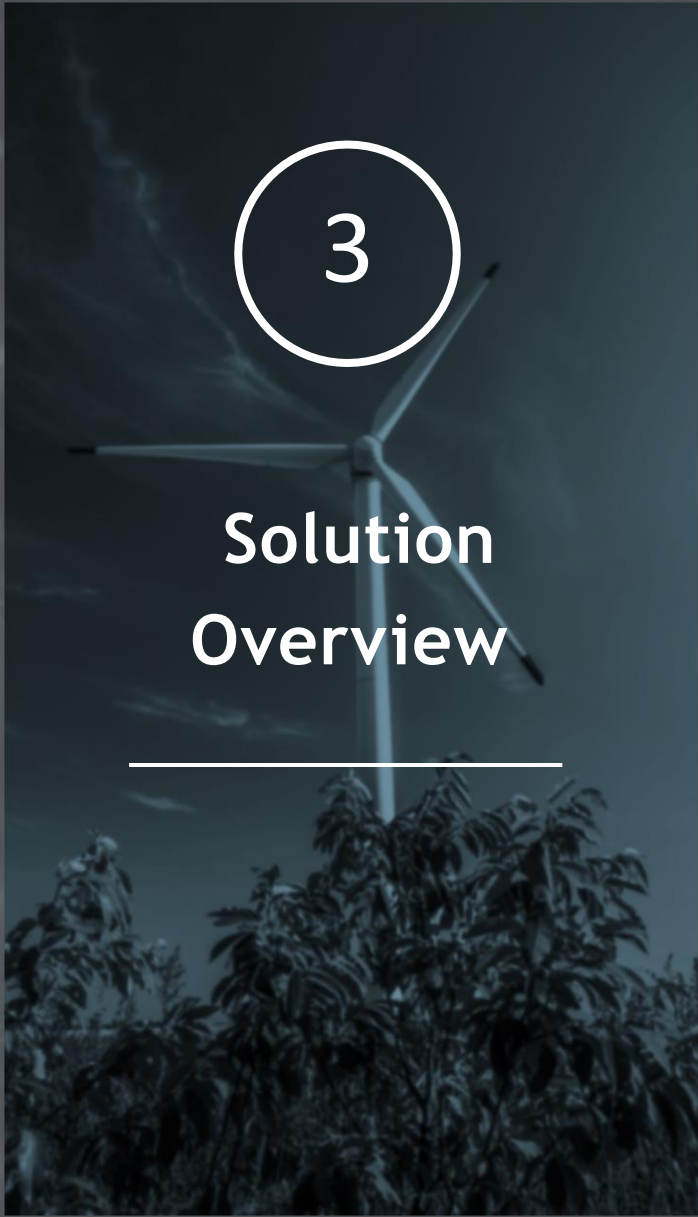
1

Company and Case Overview

A dark, stylized image showing two hands shaking over a laptop keyboard, symbolizing a deal or agreement. The image is overlaid with a white circle containing the number 2.

2

Problem Identification

A dark, stylized image of a wind turbine against a cloudy sky, with some foliage in the foreground. The image is overlaid with a white circle containing the number 3.

3

Solution Overview

Company and Case Overview: Where does MLPP stand and where does it want to pivot through launch of the decumulation product?

Maple Leaf Pension Plan

Canada's top Defined Contribution (DC) Plans

- ✓ **C\$60 billion** in Assets under Management (AUM)
- ✓ 200000+ Members
- ✓ 10 year **8.5% Net Annualized Return**

USP



Internal Portfolio Management



Cost Effective Investment Products

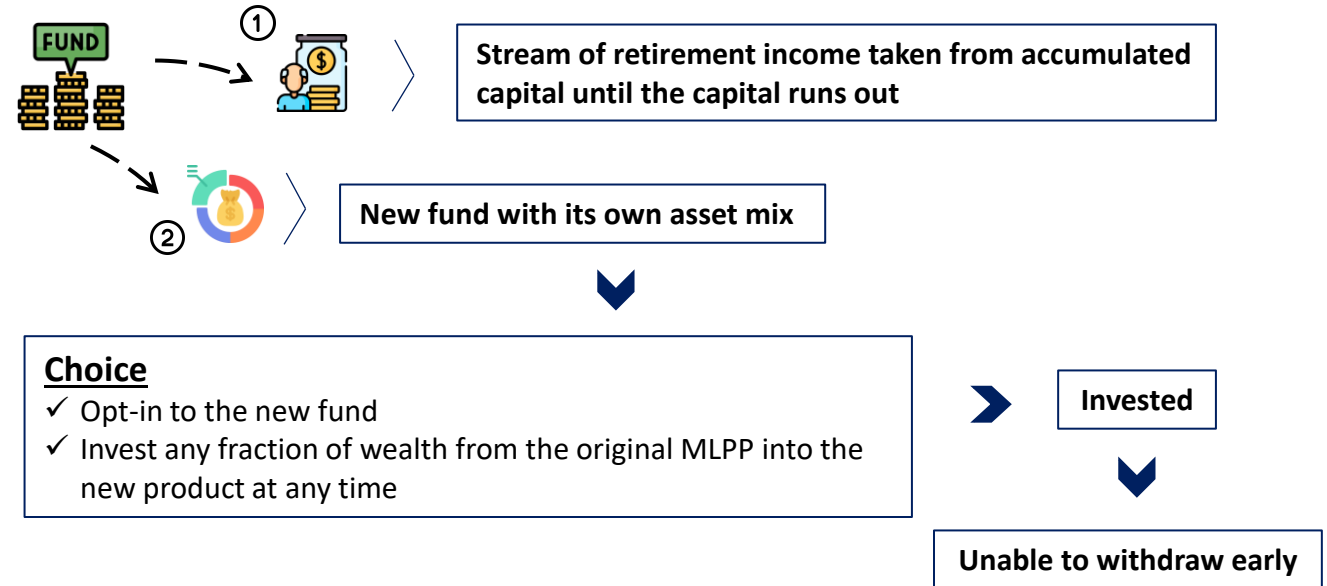
Investment Strategy

- ✓ Contributed by both members and employers
- ✓ Invest in more **illiquid** and **private assets** to seek higher risk-adjusted returns
- ✓ Investment Funds
 1. Balanced Growth Fund
 2. Bond and Cash Portfolio

Not permitted to withdraw until retirement (65 years)

Sources: Case Study, Team Analysis

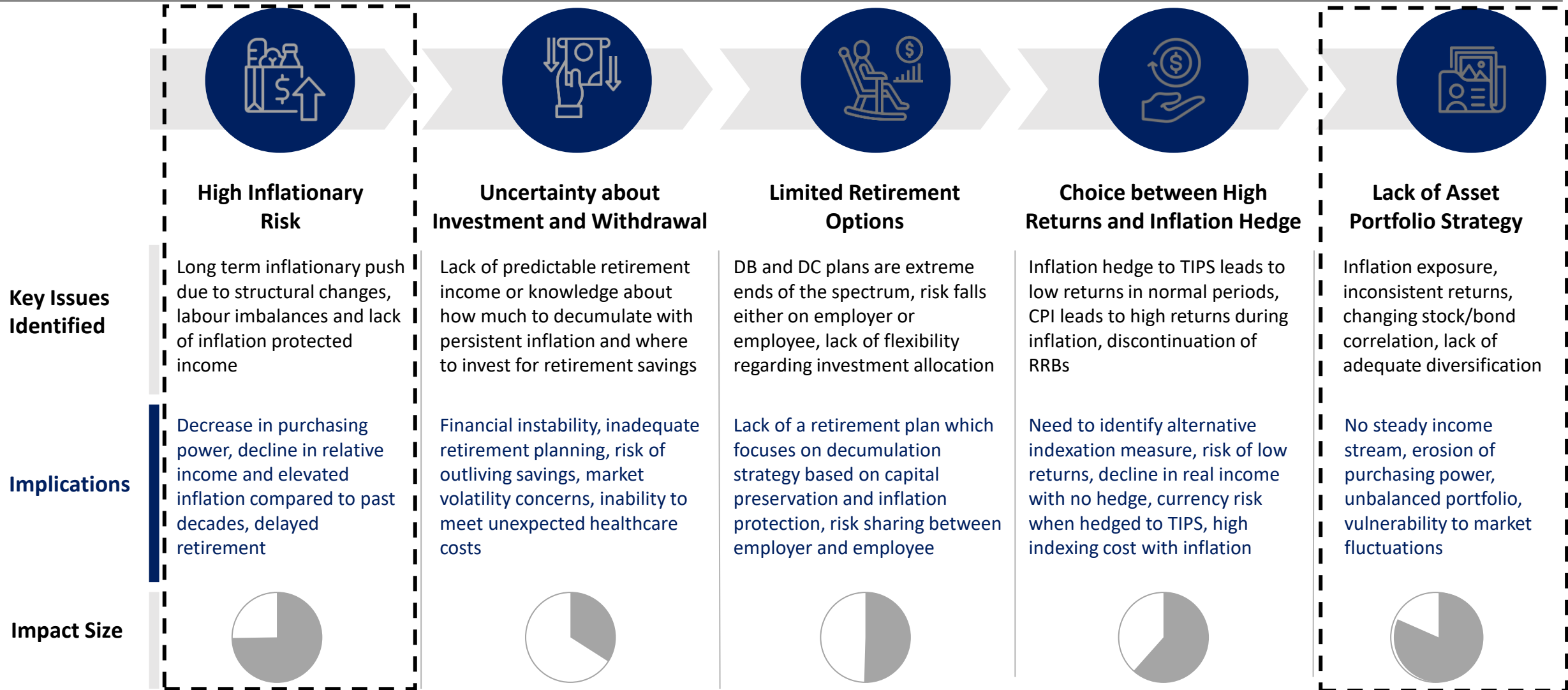
The New Decumulation Product



Key Objectives of the Fund

1. **Sustainable Retirement Income** - Ensure financial security by providing retirees a reliable stream of income every year
2. **Hedge Against Inflation** - Ensure retiree's income remains resilient against rising costs and they are able to maintain the same standard of living
3. **Diversified Investment** - Maximize returns and manage risk to provide retirees a balanced portfolio to suit their risk tolerance

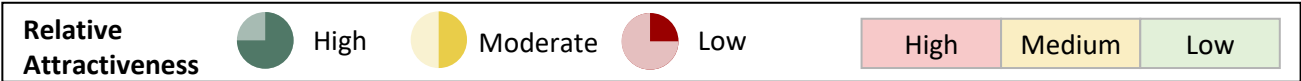
Summary (1/3): Analysing the key customer and industry problems to develop an effective decumulation and asset allocation strategy for MLPP



Sources: Team Analysis

Summary (2/3): Understanding suggested solution by targeting key problems through adoption of suitable policies and analysing their potential impact

	Complication	Proposed Solution	Parameters of Assessment				Overall Attractiveness
			Risk Involved (High/Medium/Low)	Degree of Impact	Complexity of Implementation	Duration of Implementation (Rankings 1-5: Longest - Fastest)	
1	Withdrawal Rules for Income Protection	Floor and Ceiling Limit	Low	High	High	4 th	
2	Indexation Policy: Purchasing Power	PCE Indexation	Medium	High	Low	5 th (Least Time)	
3	Indexation Policy: Balancing Returns	PCE Indexation	Medium	Low	Medium	3 rd	
4	Portfolio: Inflation Resiliency	Adjusted Returns	Low	Low	High	2 nd	
5	Portfolio: Asset Classes	Diversification	High	Medium	Medium	1 st (Most Time)	



Sources: Team Analysis

Summary (3/3): 3 Pronged Solution - Accumulation, Risk-Return and Decumulation can lead to a strong portfolio which battles inflation

Strategy Overview

1

Accumulation Phase: Assumed that automation industry workers are saving 20% of their average annual salary and giving it to the pension fund. Based on that we have devised the total investment made by them over the course of their working life.

2

Risk-Return: We have developed a diversified portfolio with private market exposure and public market exposure to result in lower risk and higher returns. Based on that we have calculated the final portfolio value. Our returns are PCE inflation adjusted

3

Decumulation Phase: Our withdrawal strategy includes a floor and ceiling amount and annual sum drawings to meet the life of people so that they don't run out of cash at any point of their retirement. in



Key Financials

3.30%

Sharpe Ratio

95%

Adaption Rate

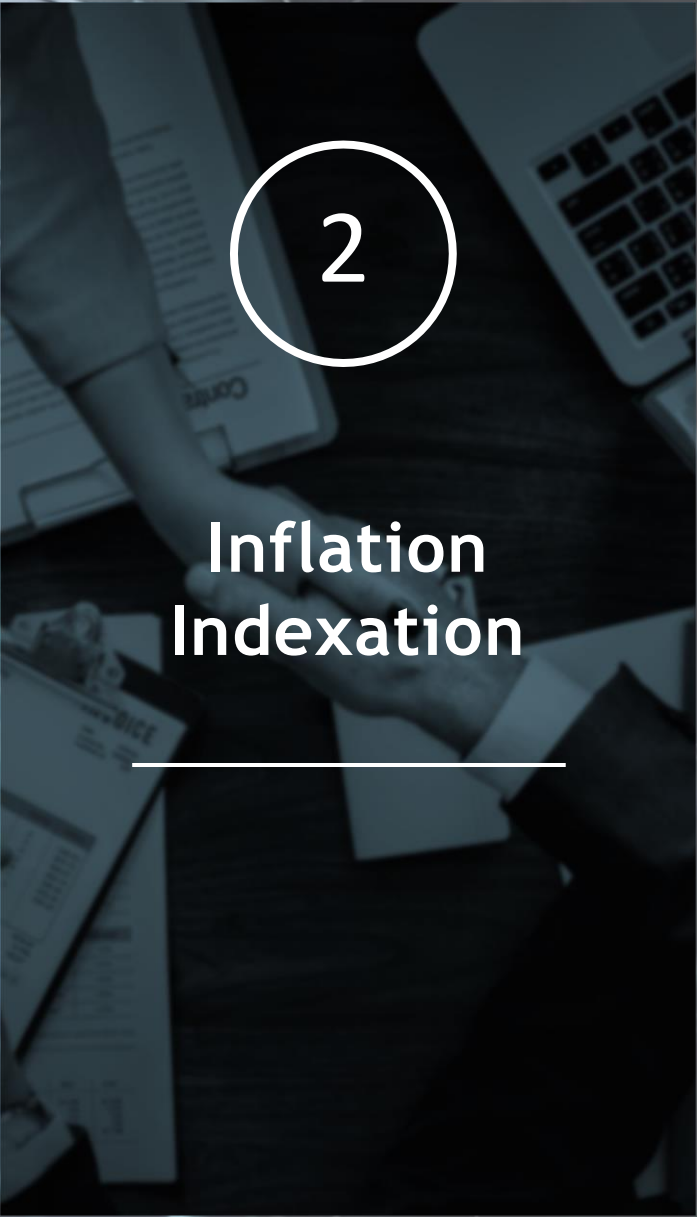
6%

Portfolio Returns

A dark, semi-transparent map of Europe is the background for the first panel. The map shows various countries and cities, with a focus on Eastern Europe. A white circle with the number 1 is centered in the upper half of the panel.

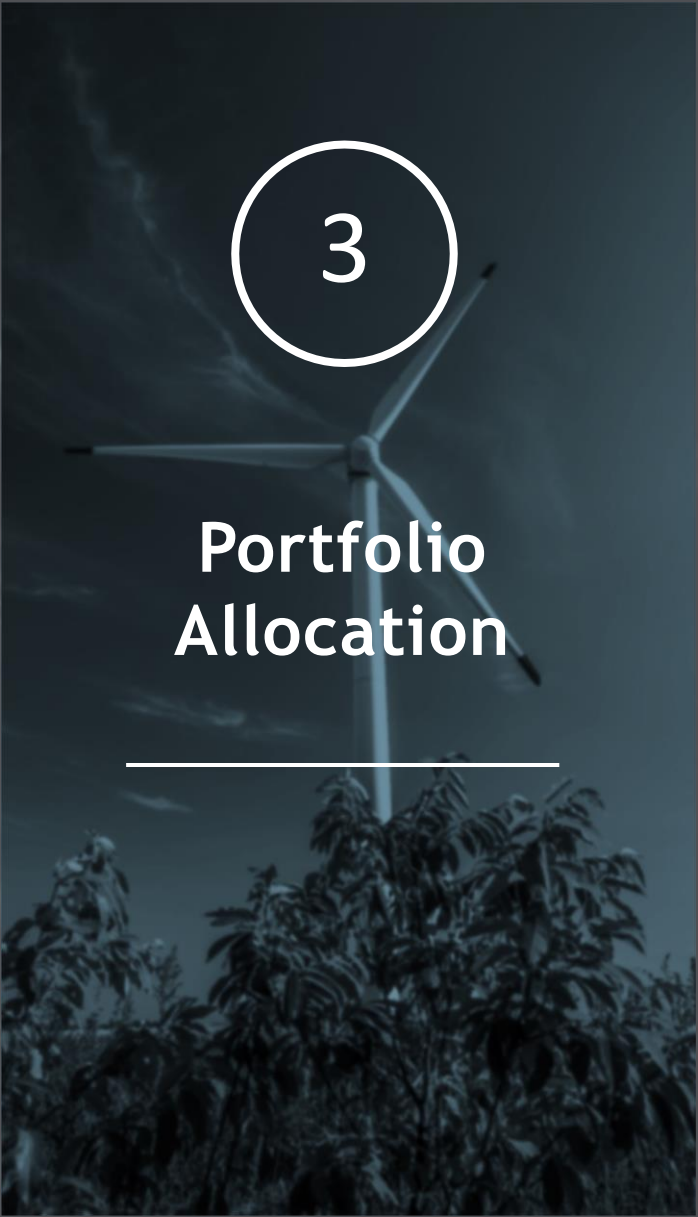
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Withdrawal Strategy

A dark, semi-transparent image of two hands shaking over a desk with papers and a laptop is the background for the second panel. A white circle with the number 2 is centered in the upper half of the panel.

2

Inflation Indexation

A dark, semi-transparent image of a wind turbine against a cloudy sky is the background for the third panel. A white circle with the number 3 is centered in the upper half of the panel.

3

Portfolio Allocation

Conducting an in-depth analysis of traditional withdrawal strategies to formulate an ideal decumulation strategy for MLPP

	Dollar Plus Inflation Strategy	Percentage of Portfolio Strategy
Definition	It involves spending a percentage of your portfolio the first year and adjust that amount in subsequent years based on inflation . Example: The 4% rule	It involves an annual withdrawal of a fixed percentage from the retiree’s portfolio. With an aim to provide a sustainable income stream while adjusting to market fluctuations and minimizing the risk of outliving one's assets.
Usability	Good to use if the retiree’s main priority is maintaining a steady level of spending.	Good to use when primary purpose is to ensure that retiree’s entire savings are not depleted.
Benefits	✓ Maintains Purchasing Power ✓ Simplified Withdrawal Management ✓ Flexibility in Asset Allocation	✓ Longevity Risk Management ✓ Dynamic Income Adjustments ✓ Market Fluctuation Responsiveness
Associated Risks	✓ Longevity Risk ✓ Inflation Risk ✓ Market Volatility	✓ Variable Income Levels ✓ Inflation Risk ✓ Market Volatility

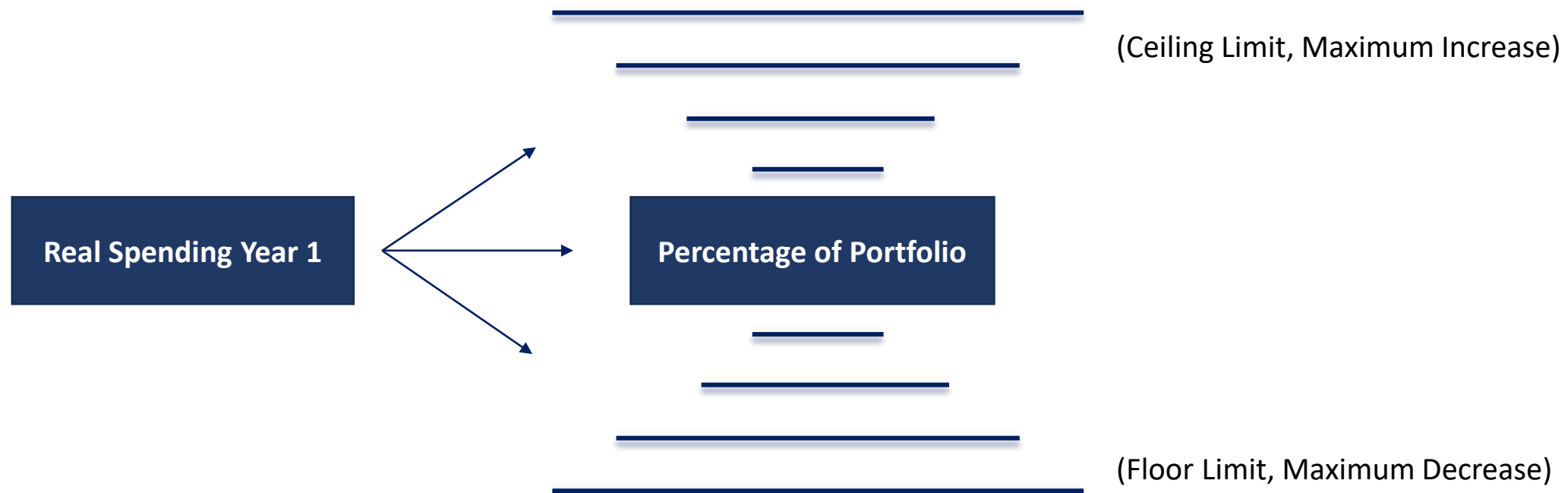
Sources: Team Analysis

KIPS Ideal Decumulation Strategy: 4% Original Withdrawal Rate, subject to 100% PCE Indexation and Floor and Ceiling Withdrawal limit

Dynamic Spending Strategy

Dynamic spending provides retirees the stability of the dollar plus inflation plan and the flexibility in spending of the percentage of portfolio plan

(Hybrid of Dollar Plus Inflation and Percentage of Portfolio Strategies - Cover current spending while preserving portfolio)



Sources: Team Analysis

Example: Working of the dynamic spending withdrawal strategy from an individual standpoint

Accounting for Dynamic Spending

Inflation not accounted for, refer Decumulation Model

Varying Floor and Ceiling Rates across Cases

Arriving at Available Amount for the Current Year

Particulars	Year 1
Portfolio Value at Retirement	1,000,000
Comfortable Withdrawal Rate	4%
First Year Decumulation	40,000
Floor Percentage	Less 2.5%
Ceiling Percentage	Add 5%

Particulars	Year 2	Year 3	Year 4
Portfolio Value at the Beginning of the year	\$10,10,000.00	\$9,29,000.00	\$11,80,000.00
Is the Portfolio Value Higher or Lower than 12 months ago?	Higher	Lower	Higher
Annual Spending of Previous Year	\$39,750.00	\$40,400.00	\$39,390.00
4% of current portfolio value	\$40,400.00	\$37,160.00	\$47,200.00
Floor (Previous year's spending - 2.5%)	\$38,756.25	\$39,390.00	\$38,405.25
Ceiling (Previous year's spending + 5%)	\$41,737.50	\$42,420.00	\$41,359.50
Amount available for current year spending/ Withdrawal Amount	\$40,400.00	\$39,390.00	\$41,359.50
Remarks	(here stick to Current Portfolio Rate as it is between Ceiling and Floor)	(here, use floor value because it is not as extreme as the proposed withdrawal rate)	(here, use ceiling value because it is not as extreme as the proposed withdrawal rate)

Sources: Team Analysis

The working of the KIPS proposed withdrawal model based on dynamic spending and indexation, with **withdrawal limits** across 30 years

Model Inputs	
Retirement Age	65
Initial Decumulation Fund Balance	\$20,00,000
Average Portfolio Return	6%
Volatility Portfolio Return	2%
Average Inflation (Norm. Dist.)	2.25%
Volatility Inflation (Norm. Dist.)	0.99%

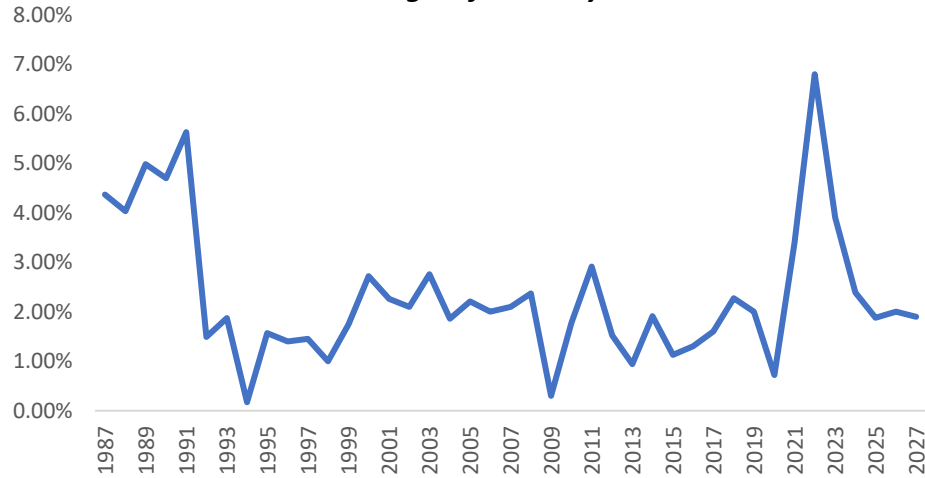
Year	Inflation	Forecasted YoY Portfolio Return	Portfolio Value at the Beginning of Year	Is the Portfolio Value Higher or Lower than 12 months ago?	Annual Spending Previous Year	4% of Current Portfolio Value (Fully Indexed)	Indexed Floor Amount (Previous Year's Spending - 2.5%)	Indexed Ceiling (Previous year's spending + 5%)	Amount available for current year spending/ Withdrawal Amount
1	2.19%	6.35%	2000000.00		N/A	81754.88	NA	NA	81754.87597
2	0.92%	6.81%	2048837.90	Higher	81754.88	82708.26	80445.10	85843.63	82708.26
3	1.03%	7.83%	2120090.96	Higher	82708.26	85673.75	81467.96	86844.69	85673.75
4	2.55%	7.61%	2189255.04	Higher	85673.75	89806.45	85665.04	89958.46	89806.45
5	2.08%	5.10%	2206429.42	Higher	89806.45	90089.27	89378.94	94297.80	90089.27
6	1.58%	7.87%	2282807.49	Higher	90089.27	92756.27	89226.06	94594.75	92756.27
7	1.70%	4.35%	2285306.84	Higher	92756.27	92964.64	91973.18	97395.10	92964.64
8	2.67%	9.33%	2396913.11	Higher	92964.64	98436.82	93061.00	97613.90	97613.90
9	3.68%	3.52%	2379494.64	Higher	97613.90	98680.97	98674.51	102495.63	98680.97
10	1.35%	6.47%	2428413.27	Higher	98680.97	98445.62	97510.60	103616.03	98445.62
11	0.76%	9.81%	2558557.00	Higher	98445.62	103124.71	96718.30	103368.91	103124.71
12	3.95%	7.28%	2634130.04	Higher	103124.71	109530.90	104521.79	108281.99	108281.99
13	0.22%	8.42%	2737123.20	Higher	108281.99	109724.75	105806.20	113697.09	109724.75
14	2.96%	4.59%	2748005.37	Higher	109724.75	113169.16	110143.72	115212.02	113169.16
15	2.36%	11.56%	2939530.69	Higher	113169.16	120355.23	112943.09	118828.64	118828.64
16	1.88%	7.12%	3019910.92	Higher	118828.64	123072.10	118040.55	124771.09	123072.10
17	2.77%	5.56%	3057833.98	Higher	123072.10	125702.87	123320.57	129226.73	125702.87
18	2.71%	6.12%	3111450.77	Higher	125702.87	127827.75	125878.63	131989.04	127827.75
19	2.49%	3.12%	3076795.61	Lower	127827.75	126134.52	127733.57	134220.16	127733.57
20	2.37%	7.17%	3162342.88	Higher	127733.57	129489.00	127489.27	134121.28	129489.00
21	1.75%	7.63%	3264411.82	Higher	129489.00	132856.07	128455.87	135964.47	132856.07
22	1.64%	7.51%	3366769.46	Higher	132856.07	136874.01	131653.87	139499.89	136874.01
23	3.12%	9.33%	3531270.67	Higher	136874.01	145656.84	137614.91	143718.75	143718.75
24	1.56%	4.12%	3525221.98	Lower	143718.75	143209.43	142312.55	150905.70	143209.43
25	3.53%	2.37%	3462016.98	Lower	143209.43	143372.44	144561.53	150370.94	144561.53
26	3.96%	8.59%	3603875.19	Higher	144561.53	149857.45	146523.05	151790.64	149857.45
27	2.36%	6.90%	3692330.45	Higher	149857.45	151183.96	149564.36	157351.35	151183.96
28	3.13%	7.14%	3794126.50	Higher	151183.96	156520.70	152023.36	158744.19	156520.70
29	3.51%	4.82%	3813087.12	Higher	156520.70	157880.78	157967.94	164347.77	157967.94
30	0.60%	5.83%	3868274.41	Higher	157967.94	155661.17	154944.65	165867.34	155661.17

Sources: Team Analysis

The uncertain inflation landscape in Canada coupled with the concerns of retirees presses the **need for indexation** of inflation

The Current Inflation Landscape

The Volatility of long term inflation trends highlight the need to hedge inflationary risk

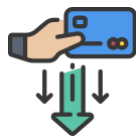


YoY Inflation Rate from 1987 to 2027

Key Drivers



71% of Canadians found saving for retirement to be more challenging due to inflationary pressures



78% are saving less for retirement primarily due to spending more on essentials



Inflation rate is **more than double** the central bank's target rate of 2%



Sticky inflation risk due to sudden rise in overnight rate to **4.75%**

The Need for Indexation

Reasons Involved..

1. Erosion of **purchasing power** of old- age benefits with time if not indexed to a suitable measure
2. Reassures retirees that their assets will keep pace with inflation with **high volatility** in Canada
3. Ensures timely retirement at a time when retirement age has risen to **65 years** in Canada



Inflation Indexation

Indexation is a method of linking the price or value of an asset to a price or price index of some type to adjust for inflation.

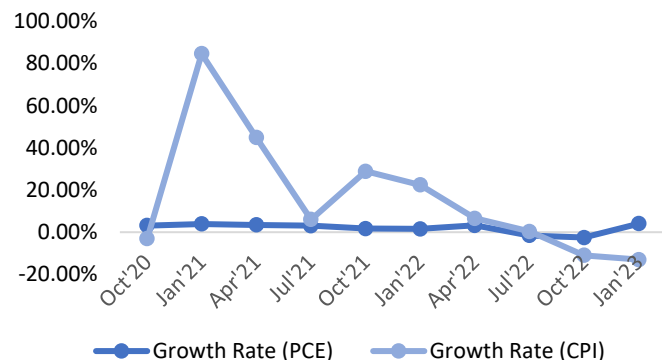
Indexing retiree's benefits in MLPP proves to be the ideal hedge:

- ✓ Easy estimation of total gain or losses incurred
- ✓ Adjustment of purchase price with inflation level
- ✓ Higher portfolio returns due to increase in purchase price

PCE Index is the ideal macroeconomic indicator for achieving inflation indexation due to its comprehensiveness and synergy with the target group

PCE vs CPI – Relative Stability

PCE vs CPI Growth Rate Variations



The **variability of the CPI** coupled with the **relative stability of the PCE** index proves the adequacy of the selection of PCE as an inflationary hedge index

Choice of Economic Indicator

Measure	Core Inflation	Target Audience Synergy	Outliers Accounted	General Acceptability
CPI	✓	✗	✗	✓
CPI Trim	✓	✗	✗	✗
PCE Index	✓	✓	✓	✓
WPI	✗	✗	✓	✓
CPI Median	✓	✓	✓	✗

Sources: IMF, Statistique Canada, CEIC Data

Personal Consumption Expenditure Price Index

Measures change in the cost of living for households by tracking the prices of a basket of goods and services, each with different weightings

Key Characteristics:

- ✓ **Comprehensiveness** - Covers government and employers that fund medical care and insurance, NPOs and imputed financial costs
- ✓ **Expenditure Diversification** – Reflects all aspects of retiree spending
- ✓ **Stability** - Smoother adjustment process for retirees' benefits
- ✓ **Flexibility** - Weights can change as people substitute away from some goods and services toward others
- ✓ **Preferred Measure** - Federal Reserve's ideal inflation measure

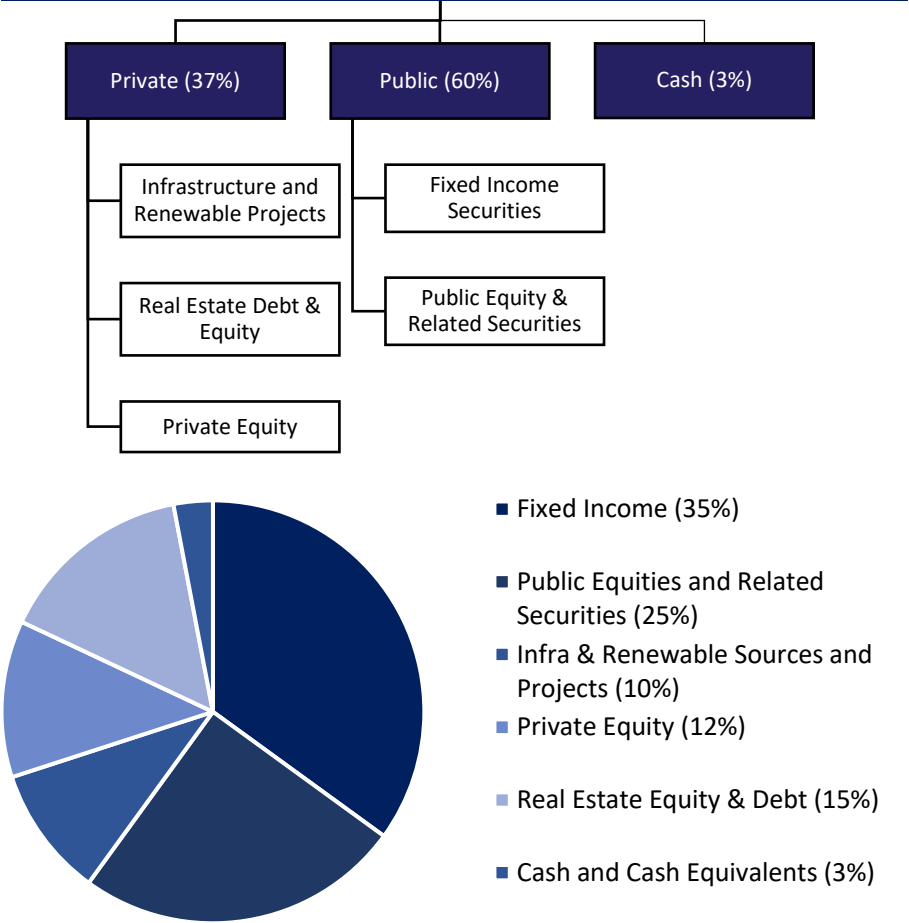
Components of the PCE Index

Components	Coverage
Medical Care	25%
Housing including Energy	17%
Recreation	13%
Transportation including Energy	15%
Food and Beverages	30%

Provides an adequate mix of components covering the expenses dominant among the **65+ age group** to ensure **inclusiveness** and accurate **projection of inflation cost**

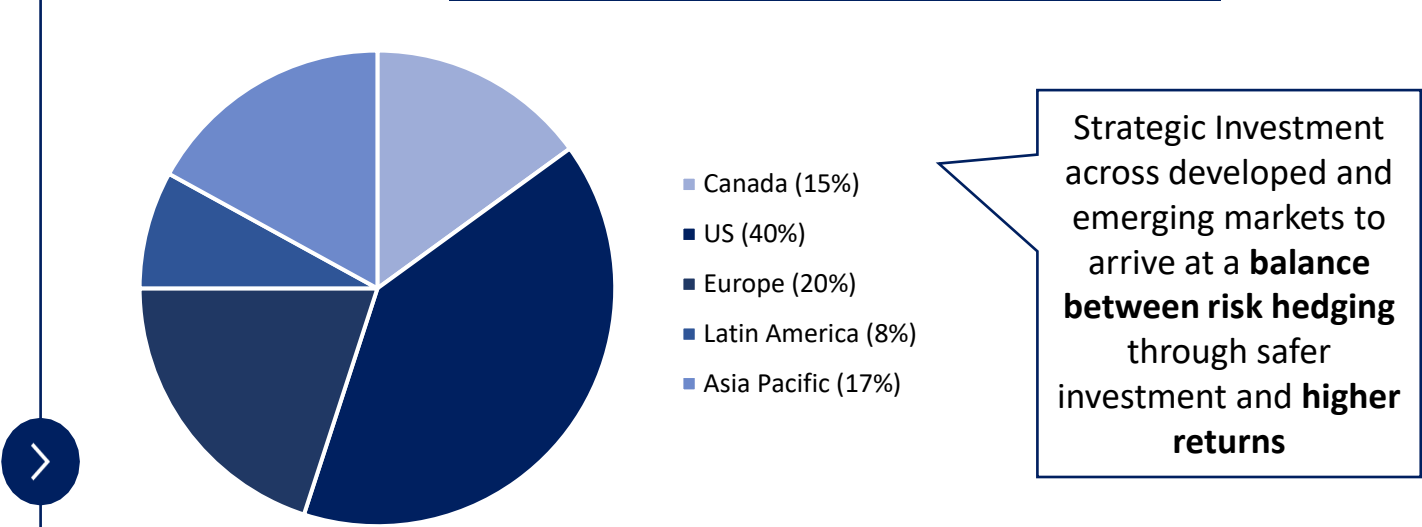
Spreading out pension premium accumulation across asset classes to optimise balance between risk and return obtained

Portfolio Allocation Across Markets and Sectors



Formulating a Diversified Portfolio

Portfolio Allocation Across Countries



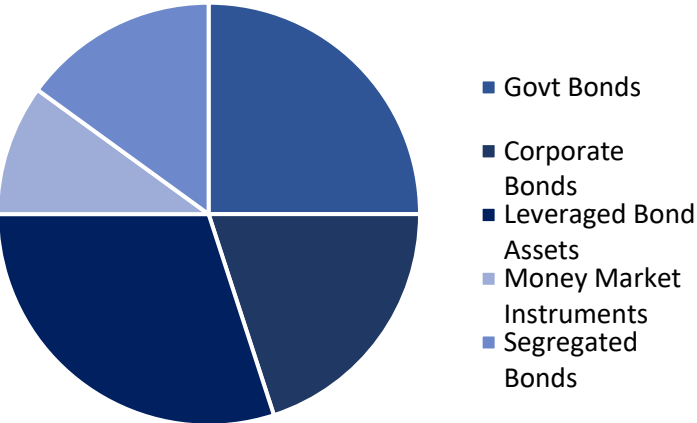
Assumption Drivers for the Portfolio

- Positive Asset class weightings, no short positions
- Reducing Home Bias : Pension Funds Transitioning away from home base
- Cash weighting <=3% to minimize cash drag
- Maximum alternatives(42%) and transition bonds(7.5%) to reduce overallocation risks
- Minimum 15% Canadian equities to promote stakeholder benefits
- No more than 5% absolute changes between each stage

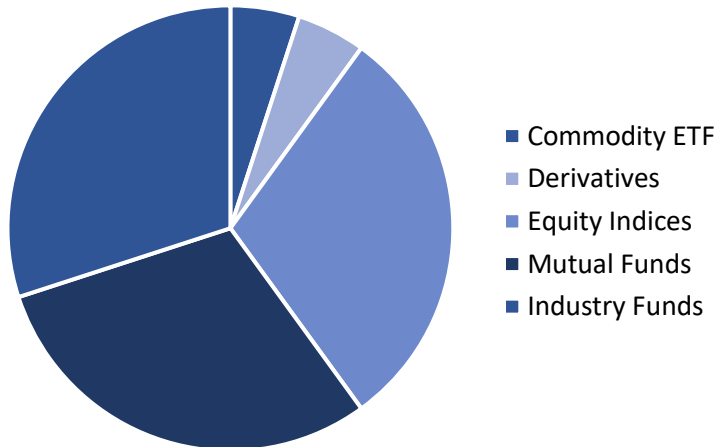
Sources: Team Analysis, Blackrock

Spreading out pension premium accumulation across asset classes to optimise balance between risk and return obtained

Fixed Income Instruments



Public Equities and Related Securities



Deep Diving into Public Market Allocation

PUBLIC MARKET ALLOCATION			
Portfolio Breakdown			
Generic Equity Breakdown :		Fixed Income :	
Equity indices	30%	Government Bonds	25%
Commodity etf	5%	Corporate Bonds	20%
Mutual funds	30%	Leveraged Bond Assets	30%
Industry funds	30%	Money Market Instruments	10%
Derivatives	5%	Segregated Bonds	15%
Segment-Wise Breakdown :		Segment-Wise Breakdown :	
Equity Indices		Government Bonds	
Canadian Equities(TSX Composite)	40%	Canadian Government Bonds, 40%	40%
US Equities(S&P 500)	30%	US Treasury Bonds, 30%	30%
Global Equities(MSCI World)	20%	International Sovereign Bonds, 20%	20%
Emerging Markets Equities(MSCI EMI)	10%	Provincial/Municipal Bonds, 10%	10%
Commodity E.T.F's		Corporate Bonds	
Energy(XLE)	20%	Investment-Grade Corporate Bon	40%
Precious Metals(GLD)	15%	High-Yield (Junk) Bonds, 20%	20%
Agriculture(DBA)	10%	Emerging Market Corporate Bonc	20%
Broad Commodity(DJP)	5%	Convertible Bonds, 10%	10%
		Preferred Shares, 10%	10%
Institutional Combined Funds		Leveraged Bond Assets	
Canada Balanced Fund	25%	Leveraged Loans, 30%	30%
Global Balanced Fund	20%	Collateralized Loan Obligations (	25%
Canadian Equity Fund	15%	Mortgage-Backed Securities (MB	20%
US Equity Fund	10%	Asset-Backed Securities (ABS),	15%
		Other Leveraged Securities, 10%	10%
Sectoral Funds		Money Market Instruments	
Technology	15%	Treasury Bills, 30%	30%
Financial Services(BFSI)	15%	Commercial Paper, 20%	20%
HealthCare	10%	Certificates of Deposit (CDs), 20%	20%
Energy	10%	Repurchase Agreements (Repos)	15%
Consumer Discretionary	10%	Short-Term Municipal Securities	10%
Industrial Sector	10%	Other Money Market Instruments,	5%
Material Sector	10%		
Consumer Staples	10%		
Telecommunications	5%		
Utilities	5%		
Derivatives		Segregated Bonds	
Equity Index Futures	10%	Infrastructure Bonds, 30%	30%
Currency Futures	10%	Green Bonds, 20%	20%
Commodity Futures	10%	Social Bonds, 20%	20%
Stock Options(Hedge)	25%	Blue Bonds (for Ocean Conserva	15%
Equity Options(Hedge)	45%	Other Segregated Bonds, 15%	15%

Key Points

5+ Fixed Income Instruments targeting capital preservation and diversification

5% Investment in Commodity ETF for hedging, portfolio stability and growth

30% Fixed Income Investment in Leveraged Bond Assets for high returns

Sources: Blackrock, Team Analysis

Overview of the proposed \$300 Million investment in Muskrat Falls Generating Station Project

\$ 5.2 Billion Bailout in 2021 by Federal Government with Loan Guarantee – Road to Recovery



Key Players:

- ✓ **Public Investors** - Federal Government of Canada, Government of Newfoundland and Labrador
- ✓ **Private Investors** - Emera, Maple Leaf Pension Fund
- ✓ **Asset Manager** - Nelcor Energy

Future Prospects for Investment

- ✓ **Return:** The project was estimated to provide a return of more than **\$30 billion** over its lifetime, now at its later stage of development
- ✓ **Environmental:** Generation of hydroelectric power, supporting both environment and economy, building brand recognition for MLPP
- ✓ **Power:** Important source of power for mining developments, citing long term prospects

Final Stage Developments - Quick Returns

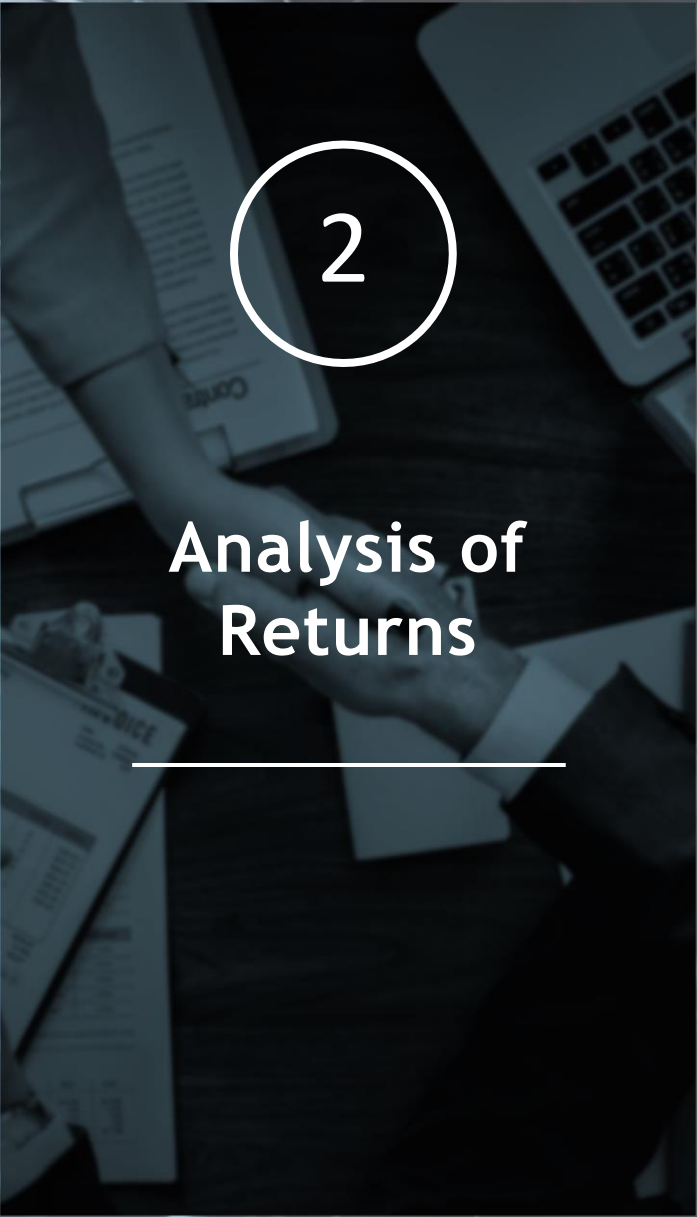
- ✓ **Final Stage Testing**
- ✓ **Transmission Lines**
- ✓ **Checks in Unit 2**
- ✓ **Work delayed due to COVID 19 to be completed**

Sources: Blackrock, Team Analysis

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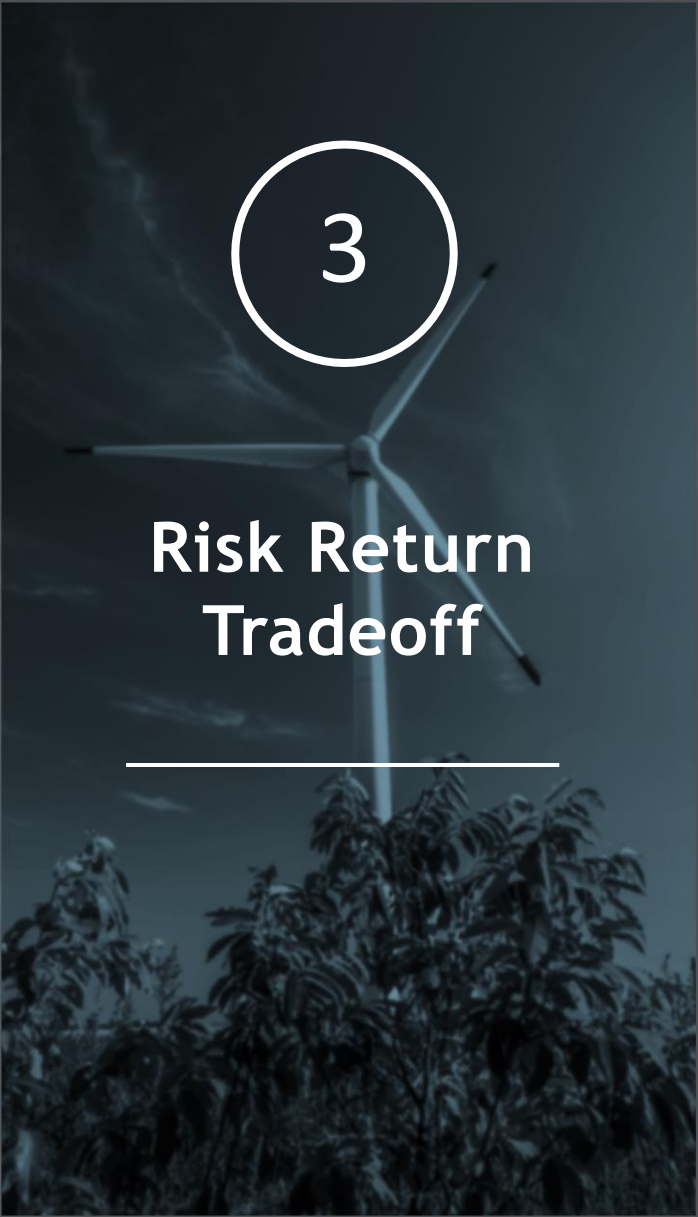
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Illustrative Example

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Analysis of Returns

A dark, stylized image of a wind turbine against a cloudy sky, with some foliage in the foreground. The image is overlaid with a white circle containing the number 3.

3

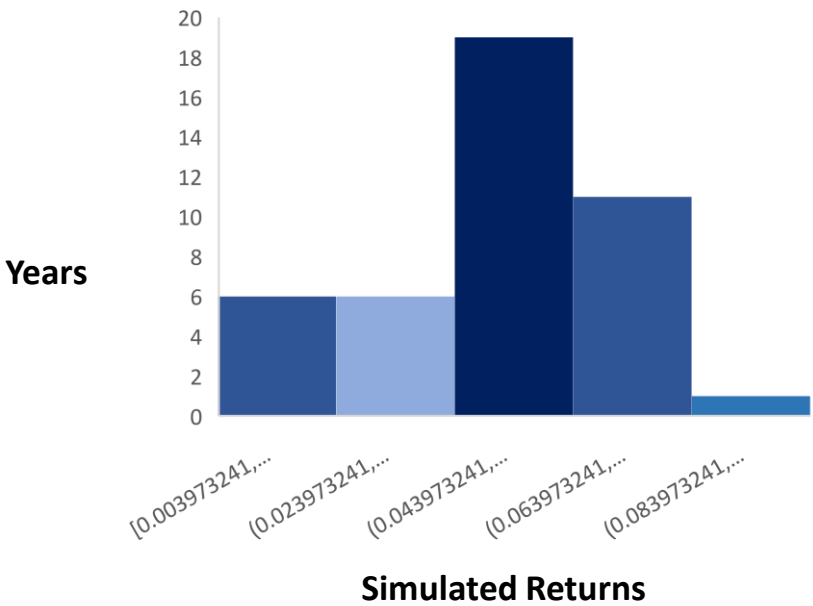
Risk Return Tradeoff

Estimated Portfolio Returns of 6% at VaR of 1.56% with the Asset Accumulation, Growth Rate of xx

Portfolio Returns Summary

Year	2024	2034	2044	2054	2064
Portfolio Statistics					
Standard Deviation	1.87	1.95	2.05	1.64	1.89
Expected Return	6.27%	6.71%	4.62%	6.19%	5.61%
Sharpe Ratio	3.69%	3.95%	2.72%	3.64%	3.30%

Value at Risk



MonteCarlo Simulation

Mean	5.18%
Std Deviation	1.99%
Min	0.40%
Max	8.33%
CMP	1000000.00

Historical Approach

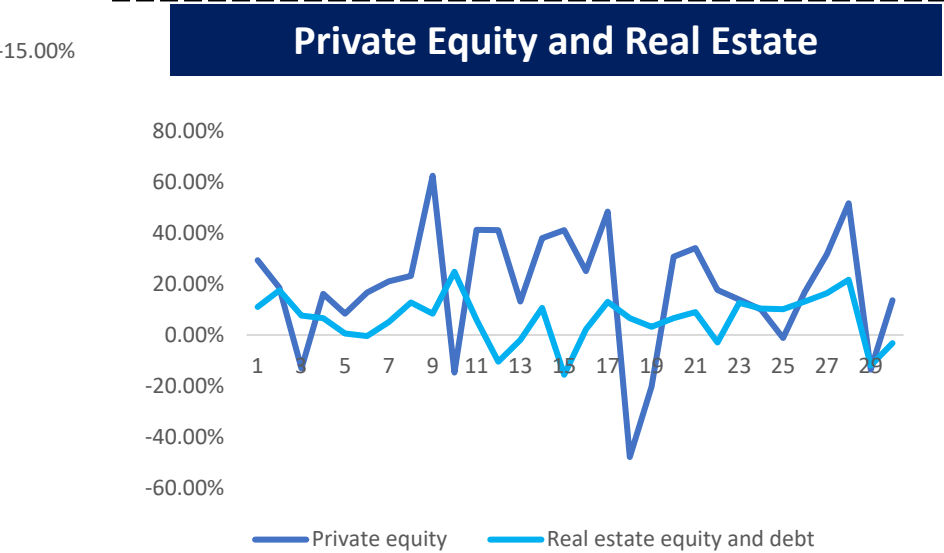
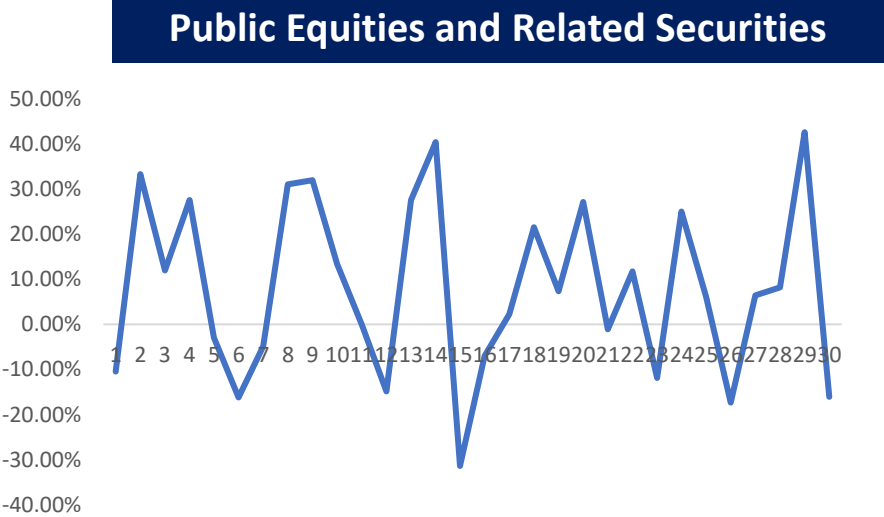
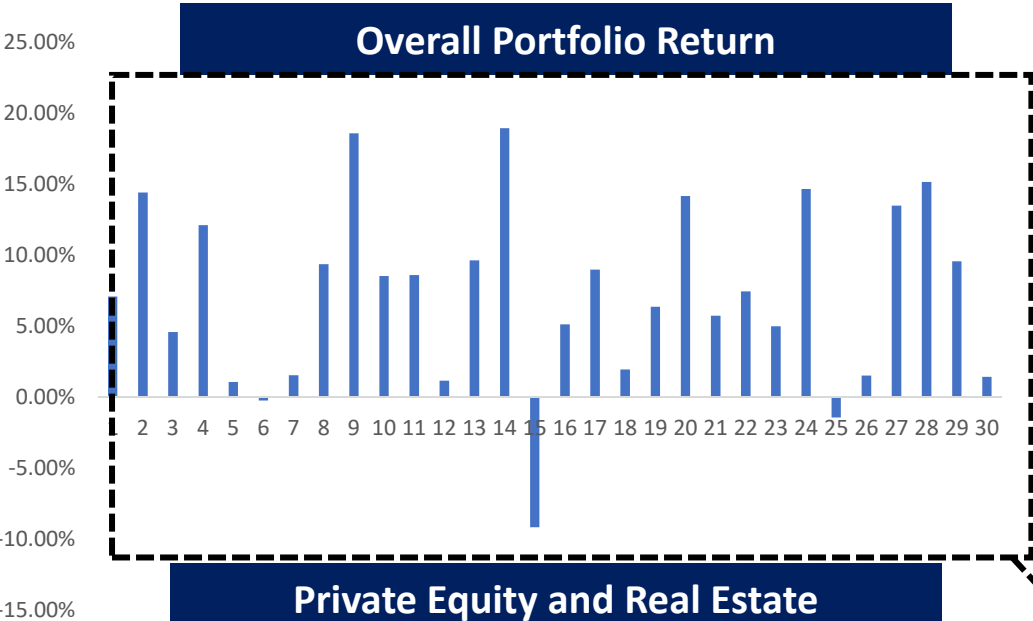
Mean	5.44%
Std Deviation	1.87%
Min	1.42%
Max	10.34%
CMP	19794.00

Percentile	Confidence	VAR(%)	Stock Price	VAR(INR)
5%	95%	1.98%	19401.17	392.83
1%	99%	0.42%	19710.13	83.87
0.50%	99.50%	0.41%	19712.81	81.19
10%	90.00%	2.33%	19333.60	460.40

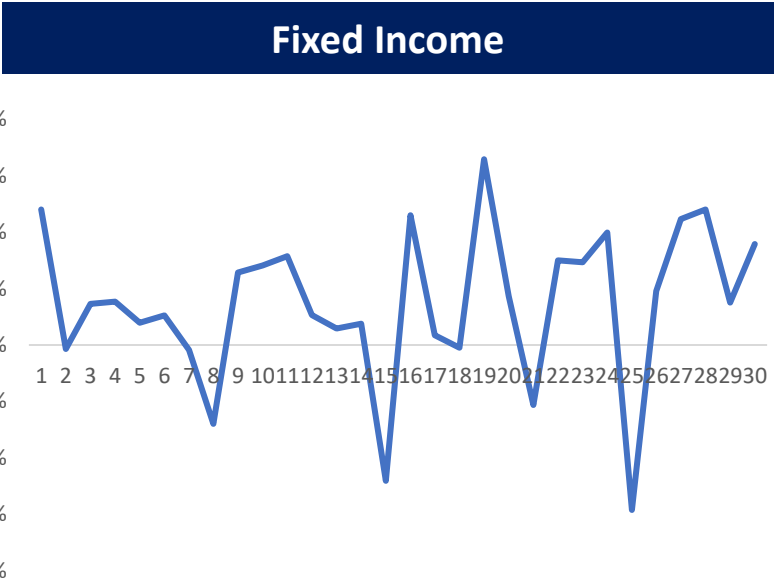
Percentile	Confidence	VAR(%)	Stock Price	VAR(INR)
5%	95%	2.53%	19292.82	501.18
1%	99%	1.56%	19484.66	309.34
0.50%	99.50%	1.49%	19498.79	295.21
10%	90.00%	3.23%	19153.86	640.14

Sources: Blackrock, Team Analysis

Analysis of Returns: Overall analysis of the returns generated through adoption of KIPS Decumulation Strategy



Net Positive
Portfolio Returns



Sources: Team Analysis

Risk - Return Tradeoff: Understanding the tradeoff concerned with the policy suggested to arrive at a conclusion

Objective	Key Questions	Recommendations	Key Questions	Recommendations
“Developing an inflationary resilient decumulation strategy by fixation of withdrawal and indexation rules and building a more comprehensive asset portfolio - James C. Brown, CEO, Maple Leaf Pension Plan	What will MLPP’s withdrawal rules look like for its decumulation product? How to ensure income protection ?	We have implemented a dynamic decumulation strategy with a floor and ceiling amount to battle inflation and uncertain scenarios. Excess returns will increase the monthly withdrawal amount, indexed to PCE.	How to strike a balance between the ideology of defined contribution plans and defined benefit plan in its new product?	Borrowed feature of DC plan in which investors can choose their investments and lock in period feature of DB plan.
	How will MLPP frame indexation policy for its decumulation product? How to protect purchasing power and balance returns?	PCE Indexation to focus only on core metrics relevant to retirees. This ensures optimal trade-off and protection of income.	How to effectively balance investment returns and inflation hedge in asset allocation portfolio?	We have effectively managed investment returns by diversifying our portfolio and exposure to private markets and hedging inflation via the PCE index.
	Should the product preserve retiree purchasing power or their relative income position ?	The product should preserve the retiree's purchasing power to keep up with inflation ultimately resulting in income stability.	What should be the degree of asset diversification for inflation resiliency in the short and long term?	We have devised five asset classes and further sub-asset classes resulting in a geographically diversified portfolio resulting in short as well as long-term gains.

Sources: Team Analysis

MIPC



KIPS Consulting

APPENDIX

Macroeconomic Trends – Forecasted Inflation

MACRO TRENDS										
Year	Inflation Rate	Annual Change			Expected Normal Inflation	annual change			Abnormal Inflationary	
1960	1.3587		average inflation rate (1960-2022)	3.741	2023	3.900		statista	7.813	
1961	1.0188	-0.34	standard deviation of inflation (1960-2022)	2.962	2024	2.390	-0.387	statista	8.908	
1962	1.0616	0.04	average annual change in inflation (1960-2022)	0.088	2025	1.880	-0.213	statista	1.890	
1963	1.6282	0.57	standard deviation of change in inflation (1960-;	1.492	2026	1.900	0.011	statista	4.137	
1964	1.9121	0.28	starting salary per annum	32.175	2027	1.988	0.046		5.525	
1965	2.3327	0.42	ending salary per annum	80.000	2028	2.075	0.044		2.119	
1966	3.8157	1.48	annual increment	2.30%	2029	2.163	0.042		3.804	
1967	3.58	-0.24	percentage investment in the fund	20.00%	2030	2.251	0.041		5.865	
1968	4.0553	0.48	average normal inflation rate (2023-2090)	4.620	2031	2.339	0.039		2.383	
1969	4.5616	0.51	standard deviation of inflation (2023-2090)	1.670	2032	2.426	0.038		3.387	
1970	3.346	-1.22	accumulation phase	2023-2065	2033	2.514	0.036		2.558	
1971	2.7049	-0.64	decumulation phase	2065-2090	2034	2.602	0.035		5.141	
1972	4.988	2.28	retirement age	60	2035	2.690	0.034		5.824	
1973	7.4876	2.5	decumulation till age	85	2036	2.777	0.033		2.821	
1974	10.9972	3.51	number of years worked	40	2037	2.865	0.032		2.909	
1975	10.6722	-0.32			2038	2.953	0.031		8.136	
1976	7.5417	-3.13			2039	3.041	0.030		4.260	
1977	7.9764	0.43			2040	3.128	0.029		5.597	
1978	8.9737	1			2041	3.216	0.028		3.260	
1979	9.1447	0.17			2042	3.304	0.027		5.010	
1980	10.1292	0.98			2043	3.392	0.027		3.435	
1981	12.4716	2.34			2044	3.479	0.026		3.523	
1982	10.769	-1.7			2045	3.567	0.025	2010	1.7769	1.48
1983	5.8636	-4.91			2046	3.655	0.025	2011	2.9121	1.14
1984	4.3048	-1.56			2047	3.743	0.024	2012	1.5157	-1.4
1985	3.962	-0.34			2048	3.830	0.023	2013	0.9383	-0.58
1986	4.1948	0.23			2049	3.918	0.023	2014	1.9066	0.97
1987	4.3561	0.16			2050	4.006	0.022	2015	1.1252	-0.78
1988	4.0282	-0.33			2051	4.094	0.022	2016	1.4288	0.3
1989	4.9836	0.96			2052	4.181	0.021	2017	1.5369	0.17
1990	4.7805	-0.2			2053	4.269	0.021	2018	2.2682	0.67
1991	5.6259	0.85			2054	4.357	0.021	2019	1.9493	-0.32
1992	1.4901	-4.14			2055	4.445	0.020	2020	0.717	-1.23
1993	1.8651	0.38			2056	4.532	0.020	2021	3.3952	2.68
1994	0.1656	-1.7			2057	4.620	0.019	2022	6.8028	3.41
1995	2.1488	1.98			2058	4.708	0.019			
1996	1.5705	-0.58			2059	4.795	0.019			
1997	1.6212	0.05			2060	4.883	0.018			
1998	0.9959	-0.63			2061	4.971	0.018			
1999	1.7348	0.74			2062	5.059	0.018			
2000	2.7194	0.98			2063	5.146	0.017			
2001	2.5251	-0.19			2064	5.234	0.017			
2002	2.2584	-0.27			2065	5.322	0.017			
2003	2.7586	0.5			2066	5.410	0.016			
2004	1.8573	-0.9			2067	5.497	0.016			
2005	2.2136	0.36			2068	5.585	0.016			
2006	2.002	-0.21			2069	5.673	0.016			
2007	2.1384	0.14			2070	5.761	0.015			
2008	2.3703	0.23			2071	5.848	0.015			
2009	0.2995	-2.07			2072	5.936	0.015			

2010	1.7769	1.48	2073	6.024	0.015	6.068
2011	2.9121	1.14	2074	6.112	0.015	6.155
2012	1.5157	-1.4	2075	6.199	0.014	6.243
2013	0.9383	-0.58	2076	6.287	0.014	6.331
2014	1.9066	0.97	2077	6.375	0.014	6.419
2015	1.1252	-0.78	2078	6.463	0.014	6.506
2016	1.4288	0.3	2079	6.550	0.014	6.594
2017	1.5369	0.17	2080	6.638	0.013	6.682
2018	2.2682	0.67	2081	6.726	0.013	6.770
2019	1.9493	-0.32	2082	6.814	0.013	6.857
2020	0.717	-1.23	2083	6.901	0.013	6.945
2021	3.3952	2.68	2084	6.989	0.013	7.033
2022	6.8028	3.41	2085	7.077	0.013	7.121
			2086	7.165	0.012	7.208
			2087	7.252	0.012	7.296
			2088	7.340	0.012	7.384
			2089	7.428	0.012	7.472
			2090	7.515	0.012	7.560

Accumulation of Funds: The 65 years time frame

Accumulation	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046	2047	2048	2049	2050	2051	2052	2053	2054	2055	2056	2057	2058	2059	2060	2061	2062	2063	2064	2065	cumulative investment
40	10,147	10,381	10,620	10,864	11,115	11,371	11,632	11,900	12,174	12,455	12,742	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3,03,240
41	10,381	10,620	10,864	11,115	11,371	11,632	11,900	12,174	12,455	12,742	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2,93,093
42	10,620	10,864	11,115	11,371	11,632	11,900	12,174	12,455	12,742	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2,82,713	
43	10,864	11,115	11,371	11,632	11,900	12,174	12,455	12,742	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2,72,093	
44	11,115	11,371	11,632	11,900	12,174	12,455	12,742	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2,61,229	
45	11,371	11,632	11,900	12,174	12,455	12,742	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2,50,114	
46	11,632	11,900	12,174	12,455	12,742	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2,38,744	
47	11,900	12,174	12,455	12,742	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2,27,111	
48	12,174	12,455	12,742	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2,15,211	
49	12,455	12,742	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2,03,037	
50	12,742	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,90,582	
51	13,035	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,77,840	
52	13,335	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,64,805	
53	13,643	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,51,470
54	13,957	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,37,827
55	14,278	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,23,870
56	14,607	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,09,592	
57	14,943	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	94,985	
58	15,288	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	80,041	
59	15,640	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	64,754	
60	16,000	16,369	16,746	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	49,114	
																																											1,17,70,081	

1,17,70,081

Basic Assumption: Case Assumption Taken

Basic Assumptions:-

1. Assuming someone starts accumulation in 2023 and is currently 20 years of age he/she will have 40 years until the age of retirement that is 60. The decumulation will start from 2063
2. A lock-in period of 20 years would mean people of age group 40-45
3. A lock in of 40 years meaning people of age group 20-25
4. Starting salary of an automotive industry worker is 32175 usd and goes up to 80000 usd. Annual increment of 2%
5. MLPP won't be able to offer decumulation products to anyone who has less than 10 years to retirement because it won't be able to generate significant returns in less than 10 yrs to sustain the entire retirement of that person

Comparative Analysis of Different Withdrawal Strategies

Parameters of Differentiation	Dollar Plus Inflation	Dynamic Spending	Percentage of Portfolio
Initial Annual Withdrawal	\$40,000	\$40000/4% of Portfolio	4% of Portfolio
Annual rise	Inflation Rate	Inflation +/- market return subject to ceiling and floor	None
Market Sensitivity	Unaffected	Responsive	Highly Responsive
Spending Stability in Short-Term	Stable	Fluctuates within limits	Unstable
Spending Flexibility	Inflexible	Flexible	Highly flexible
Portfolio Durability	Unpredictable	More Stable	Infinite

Factors Affecting Floor and Ceiling Rates

	Age		Life Expectancy		Risk Tolerance		Portfolio Composition	
	Young	Old	Higher	Lower	Higher	Lower	More Diversified	Less Diversified
Floor & Ceiling Rates	Higher	Lower	Lower	Higher	Higher	Lower	Higher	Lower

Floor & Ceiling Rates are directly proportional to Age, Risk Tolerance and Portfolio Composition and indirectly proportional to Life Expectancy

	AGE	
	Young	Old
Floor Rate	3%	2%
Ceiling Rate	8%	5%

	Risk Tolerance	
	Conservative Investor	Aggressive Investor
Floor Rate	1.5%	3.5%
Ceiling Rate	5.5%	8.5%

Factors Affecting Withdrawal Rates

Fixed	Equity_20	Equity_25	Equity_30	Equity_35	Equity_40	Balanced_20	Balanced_25	Balanced_30	Balanced_35	Balanced_40
2.50%	99.20%	98.00%	98.00%	95.90%	95.40%	99.70%	99.70%	99.70%	99.70%	99.60%
3%	98.20%	96.60%	94.50%	92.60%	90.10%	99.70%	99.70%	99.70%	99.60%	99.20%
3.50%	97.60%	94.20%	91.90%	88.50%	87.20%	99.70%	99.70%	99.30%	98.10%	96.70%
4%	96.30%	90.40%	86.90%	82.70%	81.20%	99.70%	99.40%	97.70%	92.70%	88.70%
4.50%	93.70%	88.10%	79.80%	76.70%	72.90%	99.60%	98.20%	90.60%	80.20%	73.30%
5%	90.50%	82.00%	74.60%	72.20%	66.90%	99.60%	94.30%	81.90%	66.60%	55.70%
Dynamic	Equity_20	Equity_25	Equity_30	Equity_35	Equity_40	Balanced_20	Balanced_25	Balanced_30	Balanced_35	Balanced_40
2.50%	99.50%	99.30%	98.60%	98.00%	96.60%	99.60%	99.60%	99.60%	99.60%	99.60%
3%	99.30%	98.30%	96.20%	95.30%	92.50%	99.60%	99.60%	99.60%	99.60%	99.40%
3.50%	98.50%	95.60%	94.80%	92.20%	89.10%	99.60%	99.60%	99.60%	99.00%	97.10%
4%	96.90%	94.20%	89.10%	87.50%	86.00%	99.60%	99.60%	98.60%	93.21%	87.81%
4.50%	96.80%	91.30%	84.60%	80.30%	76.30%	99.60%	99.20%	92.41%	80.12%	61.54%
5%	93.20%	84.40%	78.80%	75.50%	73.00%	99.60%	95.00%	78.82%	53.65%	34.97%
5.50%	89.90%	84.10%	74.70%	67.50%	63.30%	99.10%	82.62%	52.85%	29.47%	17.98%
6%	84.40%	72.30%	67.00%	64.30%	57.10%	94.21%	61.74%	29.57%	14.49%	5.89%

No of Years of Retirement	Withdrawal Rates
25	4%-5%
30	3.5%-4.5%
35	3.5%-4%
40	3%

-  Below 85%, which is considered risky
-  Higher success rate (Balanced vs Equity)
-  Higher success rate for Balanced (Fixed vs Dynamic)

Figure 6. Sustainable withdrawal rates across different risk profiles and time horizons

Asset allocation	Pound-plus-inflation strategy Ceiling/floor: 0%/0% (Time horizon - years)			Dynamic spending strategy Ceiling/floor: 5.0%/-2.5% (Time horizon - years)		
	10yrs	20yrs	30yrs	10yrs	20yrs	30yrs
Conservative	10.0%	5.1%	3.7%	11.1%	6.3%	4.9%
Moderate	10.4%	5.9%	4.6%	11.5%	7.0%	5.8%
Aggressive	10.6%	6.6%	5.5%	11.5%	7.3%	6.1%

Notes: Rates are gross of taxes. Any tax is assumed to be paid from the withdrawn amount. Portfolio allocations are: conservative-20% stocks/80% bonds, moderate-50% stocks/50% bonds, and aggressive-80% stocks/20% bonds. Each portfolio featured a home bias of 25% towards domestic equities and 35% towards domestic bonds. The rest of the international allocations are based on global market cap for FTSE All World and Bloomberg Barclays Global Aggregate Index. Withdrawal rates were determined using data from the VCMM. See Appendix for further description of the VCMM. The withdrawal rates are based on the portfolio meeting an 85% success rate over the stated time horizons.

Source: Matlab; Vanguard Capital Markets Model.

Risk- Benefit Analysis of Traditional Withdrawal Strategies

Dollar Plus Inflation Strategy

Benefits

- **Maintaining Purchasing Power** - By adjusting withdrawals for inflation, retirees can continue to cover their expenses and maintain their desired lifestyle throughout retirement.
- **Simplified Withdrawal Management** - By establishing a fixed withdrawal percentage and adjusting it annually for inflation, retirees can easily manage their income and budget effectively.
- **Flexibility in Asset Allocation** - It enables them to strike an appropriate balance between growth and risk management.

Risks

- **Longevity Risk** - If the retiree lives longer than anticipated or experiences higher-than-expected inflation, their withdrawals may not be sufficient to cover their spending needs.
- **Inflation Risk** - While the strategy adjusts withdrawals for inflation, there is still a risk that inflation could outpace the adjustments, eroding the retiree's purchasing power over time.
- **Market Volatility** - Impacts portfolio values, making it more difficult for retirees to maintain their desired withdrawal rate without depleting their savings too quickly.



Percentage of Portfolio Strategy

Benefits

- **Longevity Risk Management** - By withdrawing a fixed percentage of the remaining portfolio value each year, the retiree is less likely to outlive their assets, as their income adjusts to the remaining portfolio balance.
- **Dynamic Income Adjustments** - It provides a natural hedge against market fluctuations as income will increase during periods of strong investment performance and decrease during periods of poor performance.
- **Market Fluctuation Responsiveness** - Retirees to benefit from market gains while also reducing the impact of market downturns on their income.

Risks

- **Variable Income Levels** - Since the annual withdrawal amount depends on the remaining portfolio value, the retiree's income may fluctuate significantly, making it difficult to plan and budget for expenses.
- **Inflation Risk** - If the withdrawal percentage is too low, the retiree may struggle to maintain their desired standard of living in the face of rising costs.
- **Market Volatility** - Significant market fluctuations can result in substantial changes in annual income, making it difficult for retirees to plan and budget for their expenses.

MLPP – Portfolio Allocation

PENSION FUND - PORTFOLIO ALLOCATION

PORTFOLIO DIVIDE		FIXED INCOME		GEOGRAPHICAL SEGMENTATION		PUBLIC MARKET PRODUCTS	
Public Market		govt bonds	25%	canada	15%	equity indices	30%
Fixed Income	35%	corporate bonds	20%	US	40%	commodity etf	5%
Public equities and related securities	25%	leveraged bond assets	30%	europa	20%	mutual funds	30%
		money market instruments	10%	Latin America	8%	industry funds	30%
Private Market		segregated bonds	15%	asia pacific	17%	derivatives	5%
Infra & renewable sources and projects	10%						
Private equity	12%	Total	100%	total	100%	total	100%
Real estate equity and debt	15%						
cash and cash equivalents	3%						
total	100%						

ASSUMPTION DRIVERS

Positive Asset class weightings, no short positions

Reducing Home Bias : Pension Funds Transitioning away from home base

Cash weighting <=3% to minimize cash drag

Maximum alternatives(42%) and transition bonds(7.5%) to reduce overallocation risks

Minimum 15% Canadian equities to promote stakeholder benefits

No more than 5% absolute changes between each stage

MLPP – Portfolio Allocation: Public Market Allocation

PUBLIC MARKET ALLOCATION			
Portfolio Breakdown			
Generic Equity Breakdown :		Fixed Income :	
Equity indices	30%	Government Bonds	25%
Commodity etf	5%	Corporate Bonds	20%
Mutual funds	30%	Leveraged Bond Assets	30%
Industry funds	30%	Money Market Instruments	10%
Derivatives	5%	Segregated Bonds	15%
Segment-Wise Breakdown :		Segment-Wise Breakdown :	
Equity Indices		Government Bonds	
Canadian Equities(TSX Composite)	40%	Canadian Government Bonds, 40%	40%
US Equities(S&P 500)	30%	US Treasury Bonds, 30%	30%
Global Equities(MSCI World)	20%	International Sovereign Bonds, 20%	20%
Emerging Markets Equities(MSCI EMI)	10%	Provincial/Municipal Bonds, 10%	10%
Commodity E.T.F's		Corporate Bonds	
Energy(XLE)	20%	Investment-Grade Corporate Bon	40%
Precious Metals(GLD)	15%	High-Yield (Junk) Bonds, 20%	20%
Agriculture(DBA)	10%	Emerging Market Corporate Bonc	20%
Broad Commodity(DJP)	5%	Convertible Bonds, 10%	10%
		Preferred Shares, 10%	10%
Institutional Combined Funds		Leveraged Bond Assets	
Canada Balanced Fund	25%	Leveraged Loans, 30%	30%
Global Balanced Fund	20%	Collateralized Loan Obligations (	25%
Canadian Equity Fund	15%	Mortgage-Backed Securities (MB	20%
US Equity Fund	10%	Asset-Backed Securities (ABS),	15%
		Other Leveraged Securities, 10%	10%
Sectoral Funds		Money Market Instruments	
Technology	15%	Treasury Bills, 30%	30%
Financial Services(BFSI)	15%	Commercial Paper, 20%	20%
HealthCare	10%	Certificates of Deposit (CDs), 20%	20%
Energy	10%	Repurchase Agreements (Repos)	15%
Consumer Discretionary	10%	Short-Term Municipal Securities	10%
Industrial Sector	10%	Other Money Market Instruments,	5%
Material Sector	10%		
Consumer Staples	10%		
Telecommunications	5%		
Utilities	5%		
Derivatives		Segregated Bonds	
Equity Index Futures	10%	Infrastructure Bonds, 30%	30%
Currency Futures	10%	Green Bonds, 20%	20%
Commodity Futures	10%	Social Bonds, 20%	20%
Stock Options(Hedge)	25%	Blue Bonds (for Ocean Conserve	15%
Equity Options(Hedge)	45%	Other Segregated Bonds, 15%	15%

MLPP – Portfolio Allocation: Private Market

PRIVATE MARKET ALLOCATION	
Portfolio Breakdown	
Generic Breakdown :	
Infra & renewable sources and projects	10%
Private equity	12%
Real estate equity and debt	15%
Segment-Wise Breakdown :	
Infrastructure & Renewable Sources and Projects	
Renewable Energy Projects (Solar, Wind, Hydro)	40%
Transportation Infrastructure (Highways, Airports)	25%
Utilities (Water, Gas, Electric)	20%
Telecommunications Infrastructure:	10%
Other Infrastructure Projects	5%
Private Equity	
Growth Stage Startups	30%
Buyouts (Mature Companies)	25%
Venture Capital	20%
Distressed Assets/Turnarounds	15%
Mezzanine/Debt Financing	10%
Real Estate	
Commercial Real Estate (Office, Retail)	30%
Residential Real Estate (Apartments, Housing)	25%
Industrial and Logistics Properties	20%
Real Estate Development Projects	15%
Real Estate Debt Instruments (Mortgages, Loans)	10%

MLPP – Portfolio Returns

Year	Fixed Income	Public Equities and Related Securities	Infra & Renewable Sources and Projects	Private Equity	Real Estate Equity and Debt	Portfolio Return
1	12.01%	-10.43%	3.07%	29.36%	10.97%	7.07%
2	-0.35%	33.28%	13.45%	18.48%	17.51%	14.39%
3	3.66%	11.99%	7.30%	-13.14%	7.56%	4.56%
4	3.86%	27.58%	9.34%	16.06%	6.50%	12.08%
5	1.97%	-2.94%	0.18%	8.28%	0.53%	1.05%
6	2.62%	-16.15%	9.62%	16.55%	-0.47%	-0.24%
7	-0.44%	-4.86%	-3.78%	21.07%	5.05%	1.54%
8	-6.98%	31.05%	-6.50%	23.10%	12.66%	9.34%
9	6.44%	31.94%	-4.38%	62.45%	8.36%	18.55%
10	7.05%	13.37%	7.42%	-14.76%	24.82%	8.50%
11	7.87%	-0.03%	0.05%	41.21%	5.94%	8.59%
12	2.66%	-14.83%	5.82%	41.11%	-10.59%	1.15%
13	1.47%	27.58%	9.39%	13.08%	-1.98%	9.62%
14	1.88%	40.36%	20.18%	38.03%	10.55%	18.91%
15	-12.02%	-31.33%	2.96%	41.07%	-15.68%	-9.17%
16	11.51%	-6.99%	-4.78%	25.03%	2.09%	5.12%
17	0.86%	2.28%	3.49%	48.35%	12.97%	8.97%
18	-0.23%	21.50%	14.25%	-48.03%	6.59%	1.94%
19	16.47%	7.34%	7.35%	-20.32%	3.10%	6.36%
20	4.32%	27.17%	11.96%	30.64%	6.50%	14.15%
21	-5.32%	-1.08%	24.21%	34.09%	8.88%	5.71%
22	7.51%	11.72%	2.14%	17.56%	-3.05%	7.42%
23	7.34%	-11.86%	18.17%	13.79%	12.59%	4.97%
24	9.98%	25.00%	21.69%	9.95%	10.27%	14.65%
25	-14.63%	6.14%	7.77%	-1.25%	10.11%	-1.44%
26	4.78%	-17.29%	1.74%	16.87%	13.05%	1.50%
27	11.18%	6.40%	16.95%	31.69%	16.37%	13.47%
28	12.01%	8.22%	-5.82%	51.72%	21.63%	15.13%
29	3.76%	42.56%	10.23%	-13.62%	-12.01%	9.55%
30	8.96%	-16.04%	11.46%	13.57%	-3.31%	1.41%

Sources: Team Analysis

BlackRock Capital Market Assumptions			Expected returns										Lower interquartile range (25th percentile)										Upper interquartile range (25th percentile)										Lower mean uncertainty										Upper mean uncertainty										Volatili		Correlation	
Current Asset class	Asset	Index	5 year	7 year	10 year	15 year	20 year	25 year	30 year	5 year	7 year	10 year	15 year	20 year	25 year	30 year	5 year	7 year	10 year	15 year	20 year	25 year	30 year	5 year	7 year	10 year	15 year	20 year	25 year	30 year	5 year	7 year	10 year	15 year	20 year	25 year	30 year	Government	Equities																	
AUD	Private markets	U.S. direct lending	BlackRock Proxy	11.87%	11.72%	11.37%	10.88%	10.55%	10.32%	10.16%	6.65%	7.07%	6.83%	6.64%	6.50%	6.43%	6.58%	17.28%	16.63%	16.04%	15.33%	14.74%	14.24%	13.88%	8.06%	8.10%	7.60%	7.27%	7.13%	6.93%	7.05%	15.82%	15.47%	15.28%	14.62%	14.07%	13.76%	13.37%	0.118	-0.245	0.645															
CAD	Equities	Canada large cap equities	MSCI Canada Index	7.32%	7.24%	7.03%	6.90%	6.73%	6.72%	6.68%	1.25%	1.54%	1.39%	1.41%	1.33%	1.51%	1.70%	13.86%	13.34%	13.03%	12.88%	12.58%	12.22%	11.83%	3.13%	2.67%	1.93%	1.77%	1.84%	2.02%	2.12%	11.68%	12.02%	12.50%	12.25%	11.98%	11.64%	11.43%	0.143	-0.129	0.677															
CAD	Equities	China A share	MSCI China A Inclusion NET Index	11.31%	11.05%	10.44%	9.96%	9.30%	9.10%	8.74%	8.50%	-2.68%	-2.91%	-4.70%	-6.51%	-6.83%	-6.37%	-5.83%	23.28%	26.94%	27.93%	28.17%	27.52%	26.11%	25.19%	0.60%	-0.37%	-2.00%	-4.15%	-4.84%	-5.53%	-4.38%	23.61%	24.54%	25.38%	25.20%	25.17%	23.83%	0.234	-0.015	0.248															
CAD	Equities	China-Broad market equities	MSCI Emerging - Markets in CNY	10.73%	10.78%	10.32%	11.16%	11.34%	11.41%	11.56%	-1.17%	-0.88%	-0.58%	-0.16%	0.47%	1.01%	1.62%	24.84%	24.10%	23.80%	23.82%	23.37%	22.98%	22.55%	1.34%	1.33%	1.17%	0.87%	1.44%	2.28%	2.56%	20.93%	21.10%	21.60%	22.51%	22.21%	21.48%	21.36%	0.279	-0.109	0.473															
CAD	Equities	Emerging large cap equities	MSCI Emerging - Markets Index	9.32%	9.32%	9.35%	9.43%	9.46%	9.50%	9.53%	3.20%	4.59%	5.35%	6.22%	6.14%	6.08%	6.04%	15.62%	16.24%	14.74%	14.16%	13.50%	13.09%	12.74%	6.57%	6.47%	6.70%	6.77%	6.91%	7.13%	11.92%	12.14%	11.56%	12.03%	12.09%	11.62%	11.75%	0.170	-0.068	0.485																
CAD	Equities	Europe large cap equities	MSCI Europe Index	6.99%	6.99%	6.82%	6.82%	6.82%	6.82%	6.82%	0.81%	0.81%	0.81%	0.81%	0.81%	0.81%	0.81%	6.89%	6.89%	6.89%	6.89%	6.89%	6.89%	6.89%	1.92%	1.92%	1.92%	1.92%	1.92%	1.92%	1.92%	1.92%	1.92%	1.92%	1.92%	1.92%	1.92%	1.92%	0.140	-0.022	0.300															
CAD	Equities	Global ex-Canada large cap equi	MSCI World ex-Canada Index	5.96%	6.04%	6.20%	6.48%	6.66%	6.82%	6.92%	1.45%	2.14%	2.73%	3.44%	3.96%	4.34%	4.59%	10.68%	10.11%	9.80%	9.64%	9.46%	9.30%	9.33%	4.11%	4.25%	4.32%	4.61%	4.81%	4.93%	5.21%	17.88%	17.6%	18.1%	18.38%	18.58%	18.68%	0.154	-0.034	1.000																
CAD	Equities	Global small cap equities	MSCI World Small Cap Index	5.86%	5.93%	6.14%	6.46%	6.70%	6.86%	6.98%	-0.01%	0.63%	1.23%	1.97%	2.51%	2.93%	3.31%	12.02%	11.52%	11.37%	11.61%	11.3%	10.96%	10.75%	2.55%	2.73%	2.76%	3.03%	3.3%	3.65%	3.89%	3.29%	3.24%	3.62%	10.10%	10.1%	10.1%	10.5%	10.5%	10.6%	-0.136	0.301														
CAD	Equities	U.S. large cap equities	MSCI USA Index	4.58%	4.72%	5.08%	5.64%	6.04%	6.31%	6.50%	-0.32%	0.39%	1.14%	2.13%	2.96%	3.38%	3.79%	3.75%	3.28%	3.13%	3.26%	3.32%	3.34%	3.29%	2.25%	2.43%	2.73%	3.28%	3.73%	4.03%	4.31%	6.36%	7.05%	7.49%	8.06%	8.40%	8.65%	8.74%	0.144	-0.100	0.384															
CAD	Fixed income	Canada cash	3-month cash of Canadian government	3.21%	3.15%	3.04%	2.91%	2.84%	2.80%	2.77%	3.20%	3.02%	2.80%	2.62%	2.																																									

MLPP – Portfolio Returns: Assumptions

forecast										
		2023	2024	2025	2026	2027	2028	2029	2030	
Canada large cap equities	MSCI Canada Index	6.96%	6.96%	6.96%	6.96%	6.96%	6.96%	6.96%	6.96%	
China A shares	MSCI China A Inclusion NET Index	9.82%	9.82%	9.82%	9.82%	9.82%	9.82%	9.82%	9.82%	
China-Broad market equities	MSCI Emerging - China in CNY	11.14%	11.14%	11.14%	11.14%	11.14%	11.14%	11.14%	11.14%	
Emerging large cap equities	MSCI Emerging Markets Index	9.41%	9.41%	9.41%	9.41%	9.41%	9.41%	9.41%	9.41%	
Europe large cap equities	MSCI Europe index	8.58%	8.58%	8.58%	8.58%	8.58%	8.58%	8.58%	8.58%	
Global ex-Canada large cap equities	MSCI World ex Canada Index	6.44%	6.44%	6.44%	6.44%	6.44%	6.44%	6.44%	6.44%	
Global small cap equities	MSCI World Small Cap Index	6.42%	6.42%	6.42%	6.42%	6.42%	6.42%	6.42%	6.42%	
U.S. large cap equities	MSCI USA Index	5.55%	5.55%	5.55%	5.55%	5.55%	5.55%	5.55%	5.55%	
Canada cash	3-month cash of Canadian government bond quality	2.96%	2.96%	2.96%	2.96%	2.96%	2.96%	2.96%	2.96%	
Canadian aggregate bonds	FTSE TMX Canada Universe Bond Index	3.68%	3.68%	3.68%	3.68%	3.68%	3.68%	3.68%	3.68%	
Canadian corporate bonds (10+ years)	BBG Barc Canadian Corporates Long Index	3.08%	3.08%	3.08%	3.08%	3.08%	3.08%	3.08%	3.08%	
Canadian corporate bonds (all maturities)	BBG Barc Canadian Aggregate Corporate Bond Index	4.15%	4.15%	4.15%	4.15%	4.15%	4.15%	4.15%	4.15%	
Canadian government bonds (all maturities)	Bloomberg Barclays Global Treasury Canada Index	3.26%	3.26%	3.26%	3.26%	3.26%	3.26%	3.26%	3.26%	
Canadian inflation-linked government bonds	BOAICE BofAML Canadian Governments Inflation-Linked	3.91%	3.91%	3.91%	3.91%	3.91%	3.91%	3.91%	3.91%	
Canadian long bonds	FTSE TMX Canada Long Term Overall Bond Index	2.92%	2.92%	2.92%	2.92%	2.92%	2.92%	2.92%	2.92%	
China government bonds	BBG Barc China Treasury + Policy Bank Total Return Index	3.10%	3.10%	3.10%	3.10%	3.10%	3.10%	3.10%	3.10%	
Global corporate bonds	Bloomberg Barclays Global Aggregate - Corporates Index	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%	
Global government bonds	BBG Barc Global Aggregate Treasury Index	3.45%	3.45%	3.45%	3.45%	3.45%	3.45%	3.45%	3.45%	
Global high yield	Bloomberg Barclays Global High Yield Index	5.46%	5.46%	5.46%	5.46%	5.46%	5.46%	5.46%	5.46%	
Local currency EM debt	JP Morgan GBI-EM Global Diversified Index	4.54%	4.54%	4.54%	4.54%	4.54%	4.54%	4.54%	4.54%	
U.S. bank loans	S&P/LSTA Leveraged Loan Index	6.13%	6.13%	6.13%	6.13%	6.13%	6.13%	6.13%	6.13%	
USD EM debt	JP Morgan EMBI Global Diversified Index	5.23%	5.23%	5.23%	5.23%	5.23%	5.23%	5.23%	5.23%	

Asset	Index	Expected returns										Lower interquartile range (25th percentile)										Upper interquartile range (25th percentile)										Lower mean uncertainty										Upper mean uncertainty										Volatility	Correlation
		5 year	7 year	10 year	15 year	20 year	25 year	30 year	5 year	7 year	10 year	15 year	20 year	25 year	30 year	5 year	7 year	10 year	15 year	20 year	25 year	30 year	5 year	7 year	10 year	15 year	20 year	25 year	30 year	5 year	7 year	10 year	15 year	20 year	25 year	30 year																	
China-Broad market equities	MSCI Emerging - China in CNY	10.7%	10.8%	10.9%	11.2%	11.3%	11.5%	11.6%	-1.8%	-1.0%	-0.6%	-0.2%	0.5%	1.0%	1.6%	2.4%	24.1%	23.8%	23.8%	23.4%	23.0%	22.5%	1.3%	1.3%	1.2%	0.9%	1.4%	2.3%	2.6%	21.0%	21.1%	21.6%	22.5%	22.2%	21.5%	21.4%	27.9%	-10.9%	47.3%														
Emerging large cap equities	MSCI Emerging Markets Index	9.3%	9.3%	9.3%	9.4%	9.5%	9.5%	9.5%	3.3%	4.1%	4.8%	5.5%	5.9%	6.2%	6.4%	15.6%	14.7%	14.2%	13.5%	13.1%	12.9%	12.7%	6.6%	6.7%	6.7%	6.8%	6.9%	7.1%	7.4%	12.0%	12.0%	12.1%	12.1%	11.7%	17.0%	-6.8%	68.5%																
Europe large cap equities	MSCI Europe index	9.2%	9.1%	8.8%	8.5%	8.3%	8.1%	8.1%	3.8%	4.1%	4.0%	3.7%	3.6%	3.7%	3.8%	14.3%	14.2%	13.9%	13.5%	13.1%	12.8%	12.5%	6.6%	6.0%	5.1%	4.2%	4.0%	4.0%	4.0%	11.9%	12.2%	12.7%	13.0%	12.7%	12.5%	12.2%	15.1%	-5.5%	85.5%														
Global ex-Canada large cap equities	MSCI World ex Canada Index	6.0%	6.0%	6.2%	6.5%	6.7%	6.8%	6.9%	1.5%	2.1%	2.2%	2.3%	2.4%	2.5%	2.6%	10.7%	10.1%	9.8%	9.6%	9.5%	9.4%	9.3%	4.1%	4.3%	4.3%	4.6%	4.8%	5.0%	5.2%	7.9%	7.9%	8.1%	8.4%	8.6%	8.7%	13.4%	-9.4%	100.0%															
Global small cap equities	MSCI World Small Cap Index	5.9%	5.9%	6.1%	6.5%	6.7%	6.9%	7.0%	0.0%	0.6%	1.2%	2.0%	2.5%	2.9%	3.3%	12.0%	11.5%	11.4%	11.2%	11.1%	11.0%	10.8%	2.5%	2.7%	2.8%	3.0%	3.3%	3.7%	9.3%	9.2%	9.6%	10.0%	10.2%	10.2%	10.2%	16.0%	-13.6%	90.1%															
U.S. large cap equities	MSCI USA Index	4.6%	4.7%	5.1%	5.6%	6.0%	6.3%	6.5%	-0.3%	0.4%	1.1%	2.1%	2.9%	3.4%	3.8%	9.7%	9.3%	9.1%	9.3%	9.3%	9.3%	9.3%	3.3%	3.3%	3.3%	3.3%	3.3%	3.3%	3.3%	7.0%	7.1%	7.5%	8.1%	8.4%	8.6%	8.7%	14.4%	-11.0%	98.4%														
Canada cash	3-month cash of Canadian government bond quality	3.2%	3.2%	3.0%	2.9%	2.8%	2.8%	2.8%	3.2%	3.0%	2.8%	2.6%	2.6%	2.5%	2.5%	3.2%	3.3%	3.3%	3.2%	3.1%	3.1%	3.0%	3.2%	3.0%	2.8%	2.6%	2.5%	2.5%	3.2%	3.3%	3.3%	3.2%	3.1%	3.1%	3.0%	0.0%	0.0%	0.0%															
Canadian aggregate bonds	FTSE TMX Canada Universe Bond Index	3.6%	3.6%	3.6%	3.7%	3.7%	3.8%	3.8%	2.0%	2.1%	2.1%	2.2%	2.3%	2.4%	2.5%	5.1%	5.1%	5.2%	5.3%	5.2%	5.2%	2.8%	2.6%	2.5%	2.3%	2.4%	2.5%	2.6%	4.5%	4.6%	4.8%	5.0%	5.1%	5.0%	5.0%	4.4%	81.5%	15.7%															
Canadian corporate bonds (10+ years)	BBG Barc Canadian Corporates Long Index	1.8%	2.0%	2.5%	3.2%	3.7%	4.1%	4.3%	-1.3%	-0.9%	-0.5%	0.0%	0.6%	1.1%	1.4%	5.0%	5.5%	6.5%	6.9%	7.2%	7.5%	0.0%	0.1%	0.0%	0.3%	0.7%	1.1%	1.5%	3.6%	3.9%	5.0%	6.2%	6.8%	7.1%	7.2%	8.3%	61.2%	30.7%															
Canadian corporate bonds (all maturities)	BBG Barc Canadian Aggregate Corporate Bond Index	4.3%	4.2%	4.2%	4.1%	4.1%	4.1%	4.1%	3.0%	3.0%	2.9%	2.8%	2.8%	2.8%	2.8%	5.6%	5.5%	5.4%	5.5%	5.4%	5.4%	5.3%	3.6%	3.5%	3.2%	2.9%	2.9%	2.9%	5.0%	5.0%	5.2%	5.3%	5.3%	5.3%	5.2%	3.7%	66.9%	26.3%															
Canadian government bonds (all maturities)	Bloomberg Barclays Global Treasury Canada Index	3.2%	3.2%	3.2%	3.2%	3.3%	3.3%	3.3%	1.5%	1.6%	1.7%	1.7%	1.8%	1.9%	2.0%	4.9%	4.8%	4.8%	4.8%	4.8%	4.8%	4.7%	2.2%	2.1%	2.0%	1.9%	2.0%	2.1%	2.2%	4.3%	4.3%	4.4%	4.6%	4.6%	4.5%	4.3%	85.9%	1.4%															
Canadian inflation-linked government bonds	BOAICE BofAML Canadian Governments Inflation-Linked	4.0%	3.9%	3.9%	3.9%	3.9%	3.9%	3.9%	0.7%	0.8%	0.7%	0.5%	0.7%	0.9%	1.4%	7.2%	7.2%	7.4%	7.3%	7.2%	7.0%	1.6%	1.6%	1.2%	0.8%	0.7%	0.8%	1.0%	6.4%	6.7%	7.1%	7.2%	7.0%	6.8%	7.4%	53.4%	23.7%																
Canada long bonds	FTSE TMX Canada Long Term Overall Bond Index	1.8%	1.9%	2.4%	3.0%	3.5%	3.8%	4.0%	-1.5%	-1.1%	-0.7%	-0.1%	0.4%	0.9%	1.2%	5.1%	5.5%	6.3%	6.7%	6.8%	7.0%	-0.1%	-0.3%	-0.1%	0.2%	0.7%	1.1%	1.4%	3.7%	4.2%	4.9%	5.3%	6.4%	6.6%	6.7%	8.3%	75.7%	21.8%															
China government bonds	BBG Barc China Treasury + Policy Bank Total Return Index	2.6%	2.7%	2.8%	3.1%	3.4%	3.5%	3.6%	-1.1%	-0.6%	-0.1%	0.5%	1.0%	1.3%	1.5%	6.5%	6.1%	5.9%	5.9%	5.8%	5.8%	0.6%	0.6%	1.0%	1.3%	1.5%	1.7%	2.0%	4.7%	4.7%	4.8%	5.0%	5.3%	5.3%	10.3%	26.7%	2.8%																
Global corporate bonds	Bloomberg Barclays Global Aggregate - Corporates Index	3.3%	3.3%	3.3%	3.4%	3.5%	3.6%	3.7%	2.2%	2.4%	2.6%	2.7%	2.9%	3.0%	3.1%	5.7%	5.4%	5.3%	5.3%	5.2%	5.2%	3.3%	3.3%	3.2%	3.2%	3.2%	3.3%	3.4%	4.4%	4.5%	4.7%	4.8%	4.9%	4.9%	5.2%	67.3%	22.4%																
Global government bonds	BBG Barc Global Aggregate Treasury Index	3.7%	3.6%	3.5%	3.4%	3.3%	3.3%	3.3%	2.5%	2.6%	2.6%	2.5%	2.6%	2.6%	2.6%	4.9%	4.7%	4.5%	4.3%	4.1%	4.0%	4.0%	3.3%	3.2%	3.0%	2.8%	2.8%	2.8%	4.1%	4.0%	4.1%	4.0%	3.9%	3.8%	3.8%	3.6%	100.0%	-9.4%															
Global high yield	Bloomberg Barclays Global High Yield Index	5.7%	5.6%	5.5%	5.4%	5.4%	5.3%	5.3%	2.8%	3.1%	3.2%	3.2%	3.3%	3.4%	3.5%	8.6%	8.2%	7.9%	7.6%	7.4%	7.3%	1.1%	3.8%	3.6%	3.7%	3.6%	3.6%	3.7%	3.7%	1.6%	1.5%	1.4%	1.3%	1.1%	6.5%	6.5%	1.2%	24.2%	42.3%														
Local currency EM debt	JP Morgan GBI-EM Global Diversified Index	4.8%	4.7%	4.6%	4.5%	4.4%	4.4%	4.4%	1.7%	2.0%	2.1%	2.3%	2.4%	2.5%	2.6%	7.9%	7.5%	7.1%	6.7%	6.5%	6.3%	6.2%	3.1%	3.0%	3.0%	2.9%	2.9%	3.0%	6.5%	6.5%	6.3%	6.1%	6.0%	5.9%	5.8%	8.0%	39.1%	28.2%															
U.S. bank loans	S&P/LSTA Leveraged Loan Index	7.3%	7.1%	6.6%	6.0%	5.6%	5.3%	5.1%	3.1%	3.2%	2.9%	2.5%	2.3%	2.3%	2.2%	11.5%	11.0%	10.4%	9.5%	8.9%	8.4%	8.1%	4.2%	4.0%	3.5%	3.1%	2.8%	2.7%	2.6%	10.5%	10.3%	9.7%	9.0%	8.4%	8.0%	7.7%	8.6%	-11.6%	30.2%														
USD EM debt	JP Morgan EMBI Global Diversified Index	5.2%	5.2%	5.2%	5.2%	5.2%	5.3%	5.3%	1.3%	1.6%	1.6%	1.6%	2.0%	2.2%	2.3%	9.3%	9.0%	8.9%	8.9%	8.8%	8.6%	8.5%	2.5%	2.4%	2.2%	2.1%	2.1%	2.2%	2.6%	8.1%	8.1%	8.3%	8.4%	8.5%	8.4%	8.0%	9.1%	50.4%	21.0%														

Appendix 5A

Portfolio Returns – Overall returns

annual portfolio return	7.17%	6.55%	6.18%	2.50%	4.23%	4.93%	7.00%	6.27%	7.23%	8.09%	7.50%	4.70%	3.34%	7.02%	5.67%	7.31%	8.09%	5.92%	4.94%	5.07%	5.00%	10.56%	4.65%	5.05%	2.87%	5.28%	6.09%	4.38%	2.01%	4.97%	7.92%	2.13%	7.02%	4.48%	7.44%	5.77%	4.55%	5.83%	6.49%	6.00%	6.54%	6.63%	7.18%	
standard deviation	1.76%																																											
average annual portfolio returns	5.80%																																											
initial portfolio value	454,403																																											
final portfolio value	2777964.5																																											
Total Growth	498.18%																																											
risk free rate	1.02%																																											
Sharpe Ratio	3.29%	3.85%	3.64%	3.51%	2.49%	2.90%	4.47%	3.69%	4.25%	5.11%	4.41%	2.77%	1.96%	4.13%	3.33%	4.24%	4.76%	3.48%	2.91%	2.98%	2.94%	6.21%	2.73%	2.97%	1.85%	3.11%	3.58%	2.58%	1.18%	2.92%	4.66%	1.25%	4.13%	2.64%	4.38%	3.40%	2.68%	3.43%	3.82%	3.53%	3.85%	3.90%	4.22%	

Portfolio Returns – Accumulation Phase

												overall weightage in equities / fixed income / other private sectors										overall weightage in portfolio																			
																																</									

Portfolio Returns – Accumulation Phase



						overall weightage in equities / fixed income / other private sectors				overall weightage in portfolio				2013 A2014 A2015 A2016 A2017 A2018 A2019 A2020 A2021 A2022 AMean											
Public equities and related securities	25%	equity indices	Canadian Equities(TSX Composite)	12.00%	3.00%	9.55%	7.42%	-11.09%	17.51%	6.03%	-11.64%	19.13%	2.17%	21.74%	-8.66%	5.22%									
			US Equities(S&P 500)	8.00%	2.25%	26.39%	12.39%	-6.69%	11.24%	18.42%	-7.01%	28.71%	15.29%	28.79%	-19.95%	11.36%									
			Global Equities(MSCI World)	6.00%	1.50%	21.19%	19.20%	10.55%	11.04%	7.58%	-4.38%	30.12%	6.11%	31.98%	-13.08%	12.03%									
			Emerging Markets Equities(MSCI EM)	3.00%	0.75%	-6.50%	11.53%	-4.76%	15.27%	21.07%	-10.17%	21.17%	8.66%	5.93%	-14.77%	4.74%									
		commodity etf	Energy(XLE)	2.00%	0.50%	26.25%	-8.70%	-21.47%	28.02%	-9.80%	-18.21%	11.74%	-32.51%	53.31%	64.17%	10.17%									
			Precious Metals(GLD)	1.50%	0.38%	-28.33%	-21.19%	-10.47%	8.03%	12.81%	-1.94%	17.86%	24.81%	-4.15%	-0.77%	-8.35%									
			Agriculture(DBA)	1.00%	0.25%	-13.24%	2.64%	-17.20%	-3.11%	-6.06%	-8.74%	-0.70%	-2.54%	22.37%	2.53%	-2.41%									
			Broad Commodity(GDP)	0.50%	0.13%	-11.12%	-18.61%	-28.22%	12.86%	0.74%	-13.07%	7.63%	-4.12%	31.05%	17.46%	-0.54%									
		mutual funds	Canada Balanced Fund	10.50%	2.63%	13.07%	11.94%	-7.81%	21.21%	9.58%	-7.82%	21.80%	5.27%	28.05%	-6.36%	8.89%									
			Global Balanced Fund	8.00%	2.25%	NA	10%	1.22%	3.61%	12.91%	-1.57%	17.24%	12.28%	9.52%	-14.48%	5.41%									
			Canadian Equity Fund (Dividend)	6.00%	1.50%	17.33%	7.60%	-1.55%	16.93%	8.10%	-8.62%	19.36%	-1.99%	25.50%	-2.17%	7.05%									
			US Equity Fund	4.50%	1.13%	22.89%	11.69%	-0.31%	16.44%	20.83%	-5.56%	27.28%	15.08%	29.87%	-3.23%	14.50%									
		industry funds	Technology	4.50%	1.13%	48.16%	5.78%	7.13%	8.50%	27.94%	7.66%	36.50%	52.40%	7.48%	-34.22%	16.73%									
			Financial Services(BFS)	4.50%	1.13%	26.04%	2.86%	-3.65%	11.73%	23.79%	-16.33%	24.60%	-2.99%	25.10%	-8.93%	8.22%									
			HealthCare	3.00%	0.75%	36.58%	27.19%	5.26%	1.01%	25.45%	9.58%	22.35%	17.67%	24.60%	-7.10%	16.25%									
			Energy	3.00%	0.75%	25.83%	-9.94%	-21.23%	29.19%	-2.50%	-19.96%	9.27%	-13.00%	56.23%	62.86%	9.47%									
			Consumer Discretionary	3.00%	0.75%	41.09%	14.46%	6.10%	5.59%	19.87%	1.77%	27.36%	24.58%	19.65%	-11.78%	12.87%									
			Industrial Sector	3.00%	0.75%	35.53%	14.00%	-2.66%	14.50%	18.51%	-7.82%	28.81%	12.71%	29.41%	-11.62%	13.15%									
			Material Sector	3.00%	0.75%	24.93%	5.86%	-10.18%	21.51%	23.76%	-17.48%	23.53%	19.40%	27.41%	-11.79%	10.70%									
			Consumer Staples	3.00%	0.75%	19.59%	6.42%	6.14%	1.24%	17.60%	-10.70%	23.39%	7.69%	13.69%	-6.01%	7.91%									
			Telecommunications	1.50%	0.38%	22.88%	5.44%	-1.30%	24.86%	0.56%	-6.58%	12.51%	22.47%	21.60%	-19.18%	8.34%									
			Utilities	1.50%	0.38%	13.71%	25.18%	-7.38%	17.77%	10.56%	2.70%	26.80%	-1.95%	17.98%	1.68%	10.71%									
		Currency Pairs	USD	1.75%	0.44%	3.01%	7.21%	15.83%	3.62%	-2.09%	-0.09%	2.35%	1.03%	-6.49%	3.79%	2.82%									
			Canadian Dollar	1.10%	0.28%	3.01%	7.21%	15.83%	3.62%	-2.09%	-0.09%	2.35%	1.03%	-6.49%	3.79%	2.82%									
			British Pound	0.15%	0.04%	3.01%	7.21%	15.83%	3.62%	-2.09%	-0.09%	2.35%	1.03%	-6.49%	3.79%	2.82%									
			Indian Rupee	0.20%	0.05%	3.01%	7.21%	15.83%	3.62%	-2.09%	-0.09%	2.35%	1.03%	-6.49%	3.79%	2.82%									
			Japanese Yen	0.30%	0.08%	3.01%	7.21%	15.83%	3.62%	-2.09%	-0.09%	2.35%	1.03%	-6.49%	3.79%	2.82%									
govt bonds	Canada Government Bonds	10.00%	3.00%	2.70%	2.18%	1.78%	1.73%	2.05%	2.49%	1.60%	0.69%	1.68%	1.17%	2.01%											
	US Treasury Bonds	7.50%	2.20%	3.03%	2.17%	2.27%	2.45%	2.41%	2.69%	1.62%	1.01%	1.51%	3.88%	2.30%											
	International Sovereign Bonds	5.00%	1.80%	-1.37%	-2.22%	-7.24%	0.86%	11.38%	-2.60%	3.76%	10.88%	-6.25%	-22.07%	-1.79%											
	Provincial/Municipal Bonds	2.50%	0.80%	-8.21%	1.35%	-11.36%	0.78%	0.86%	2.28%	3.05%	6.08%	-1.45%	-4.22%	-1.28%											
corporate bonds	Investment-Grade Corporate Bonds	8.00%	2.88%	5.08%	-3.32%	2.55%	3.83%	-4.42%	10.51%	6.54%	-3.77%	-16.43%	0.00%												
	High-Yield (Junk) Bonds	4.00%	1.44%	6.37%	0.48%	-7.45%	16.28%	6.05%	-1.88%	11.80%	6.98%	8.29%	-9.20%	3.83%											
	Emerging Market Corporate Bonds	4.00%	1.44%	4.17%	1.59%	9.86%	8.41%	-2.92%	14.46%	5.66%	-1.81%	-17.41%	2.45%												
	Convertible Bonds	2.00%	0.72%	14.32%	3.64%	0.47%	1.73%	6.01%	-1.69%	14.75%	22.24%	3.83%	-12.54%	5.08%											
	Prefixed Shares	2.00%	0.72%				3.87%	6.05%	-1.54%	11.43%	4.26%	2.18%	-5.77%	3.01%											
Fixed Income	leveraged bond assets	Leveraged Loans	8.00%	3.24%	-30.01%	97.34%	-13.73%	-2.52%	22.74%	-11.01%	34.74%	38.96%	-19.80%	-72.59%	3.51%										
		Collateralized Loan Obligations (CLOs)	2.50%	2.70%									0.80%	0.15%	0.51%										
		Mortgage-Backed Securities (MBS)	6.00%	2.16%	-1.95%	6.38%	1.14%	1.27%	2.49%	0.82%	6.18%	4.08%	-1.43%	-11.74%	0.72%										
		Asset-Backed Securities (ABS)	4.50%	1.62%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%										
		Other Leveraged Securities	3.00%	1.08%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%	3.60%										
	money market instruments	Treasury Bills	3.00%	1.08%	NA	NA	-0.07%	1.01%	1.76%	2.03%	0.44%	-0.02%	1.99%	1.02%											
		Commercial Paper	2.00%	0.72%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%										
		Certificates of Deposit (CDs)	2.00%	0.72%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%										
		Repurchase Agreements (Repos)	1.50%	0.54%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%										
		Short-Term Municipal Securities	1.00%	0.36%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%	3.20%										
		Other Money Market Instruments	0.50%	0.18%	3.90%	3.90%	3.90%	3.90%	3.90%	3.90%	3.90%	3.90%	3.90%	3.90%	3.90%										
		segregated bonds	Infrastructure Bonds	4.50%	1.62%	2.26%	2.26%	2.26%	2.26%	2.26%	2.26%	2.26%	2.26%	2.26%	2.26%	2.26%									
Green Bonds			3.00%	1.08%	2.25%	2.25%	2.25%	2.25%	2.25%	2.25%	2.25%	2.25%	2.25%	2.25%	2.25%										
Social Bonds	3.00%		1.08%	1.61%	1.61%	1.61%	1.61%	1.61%	1.61%	1.61%	1.61%	1.61%	1.61%	1.61%											
Blue Bonds (for Ocean Conservation)	2.25%		0.81%	3.63%	3.63%	3.63%	3.63%	3.63%	3.63%	3.63%	3.63%	3.63%	3.63%	3.63%											
Other Segregated Bonds	2.25%		0.81%	3.61%	3.61%	3.61%	3.61%	3.61%	3.61%	3.61%	3.61%	3.61%	3.61%	3.61%											
Infra & renewable sources and projects	Renewable Energy Projects (Solar, Wind, Hydro)	10.00%	3.80%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%											
	Transportation Infrastructure (Highways, Airports)	6.25%	2.38%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%											
	Utilities (Water, Gas, Electric)	5.00%	1.90%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%											
	Multi-Strategy	2.50%	0.95%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%											
	Global Macro	1.25%	0.48%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%											
Private Market	Private equity	Growth Stage Startups	10.50%	3.99%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10.00%										
		Buyouts (Mature Companies)	8.75%	3.13%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10.00%										
		Venture Capital	7.50%	2.63%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10.00%										
		Distressed Assets/Turnarounds	5.25%	2.00%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10.00%										
			Mezzanine/Debt Financing	3.50%	1.33%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10.00%									
			Commercial Real Estate (Office, Retail)	12.00%	4.56%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%									
			Residential Real Estate (Apartments, Housing)	10.00%	3.60%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%									
			Industrial and Logistics Properties	8.00%	3.04%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%									
	Real estate equity and debt	Real Estate Development Projects	6.00%	2.28%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%										
		Real Estate Debt Instruments (Mortgages, Loans)	4.00%	1.52%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%										

Portfolio Returns – Accumulation Phase

STDEV	2023 F	2024 F	2025 F	2026 F	2027 F	2028 F	2029 F	2030 F	2031 F	2032 F	2033 F	2034 F	2035 F	2036 F	2037 F	2038 F	2039 F	2040 F	2041 F	2042 F	2043 F	2044 F	2045 F	2046 F	2047 F	2048 F	2049 F	2050 F	2051 F	2052 F	2053 F	2054 F	2055 F	2056 F	2057 F	2058 F	2059 F	2060 F	2061 F	2062 F	2063 F	2064 F	2065 F	Historic Avg Returns	weighted average annual returns		
12.45%	-8.80%	27.68%	23.52%	0.33%	-1.82%	35.61%	4.53%	-10.95%	-8.10%	8.59%	5.30%	6.02%	-6.51%	16.07%	16.32%	1.41%	-2.44%	-2.51%	24.71%	13.53%	5.20%	11.81%	-1.30%	8.78%	10.01%	21.63%	-16.29%	7.37%	-8.05%	-3.77%	-1.05%	-5.50%	2.48%	36.57%	-9.38%	8.09%	-7.81%	-1.43%	3.03%	7.55%	12.42%	5.57%	15.03%	5.28%	0.16%		
16.20%	11.10%	23.05%	10.97%	0.92%	17.65%	7.84%	-18.64%	21.82%	54.44%	26.16%	1.43%	-10.62%	-21.39%	-14.72%	-12.90%	29.04%	-5.67%	-0.21%	-4.12%	15.20%	6.40%	-6.23%	17.83%	-4.66%	8.34%	8.58%	14.40%	16.16%	21.46%	28.09%	11.70%	1.95%	45.17%	30.00%	34.19%	6.20%	11.80%	10.78%	28.56%	35.44%	-2.55%	18.71%	28.08%	11.37%	0.26%		
14.21%	16.45%	4.68%	13.62%	16.57%	-6.10%	11.01%	13.18%	13.44%	-7.02%	12.60%	17.76%	12.99%	4.28%	15.20%	36.76%	21.07%	-5.50%	19.30%	18.85%	-25.47%	37.67%	18.00%	8.91%	7.40%	-8.20%	-0.68%	8.44%	1.08%	24.37%	8.52%	-18.23%	-0.27%	8.01%	5.33%	16.71%	6.21%	11.63%	17.35%	-32.44%	18.70%	7.83%	2.63%	0.02%				
13.64%	0.05%	1.81%	7.27%	11.05%	-18.99%	18.93%	7.75%	-31.14%	-12.52%	-22.23%	6.30%	32.43%	11.49%	3.99%	0.55%	2.69%	-8.62%	17.33%	11.55%	-4.43%	11.31%	-14.50%	5.29%	11.39%	-0.19%	18.87%	8.03%	18.28%	-3.00%	13.01%	-7.56%	-6.53%	15.15%	3.46%	-0.77%	-5.63%	0.16%	20.09%	11.33%	-0.48%	-10.39%	-7.64%	-18.65%	8.26%	0.06%		
32.43%	-49.84%	-10.80%	54.93%	2.34%	60.29%	7.32%	-17.06%	70.63%	12.84%	3.24%	-10.40%	-20.39%	-21.32%	19.88%	-38.57%	-15.39%	-14.34%	-7.50%	-7.46%	17.31%	60.09%	37.69%	17.31%	-56.33%	-65.44%	-22.36%	35.73%	44.77%	-3.91%	55.19%	1.36%	14.22%	-8.26%	4.99%	11.53%	19.38%	-1.85%	53.67%	-13.50%	38.00%	4.75%	26.92%	-32.76%	8.26%	0.06%		
16.83%	-19.00%	-11.48%	15.54%	-6.72%	-29.79%	-0.05%	-21.89%	2.61%	-20.14%	2.56%	-6.45%	10.53%	-21.00%	-0.94%	-27.25%	-11.39%	7.77%	-8.35%	-18.66%	4.21%	-2.62%	-17.72%	-11.96%	12.31%	8.66%	-15.47%	23.30%	16.64%	8.38%	1.44%	-35.60%	-8.22%	-7.47%	11.12%	-12.41%	5.09%	5.92%	1.46%	-13.60%	-8.12%	10.64%	13.46%	8.49%	-4.17%	-0.02%		
10.84%	-7.95%	6.71%	1.15%	-1.05%	-1.55%	-19.80%	12.94%	-18.84%	12.81%	-27.38%	17.01%	8.35%	14.06%	-4.25%	-1.64%	22.21%	-6.37%	-4.17%	-27.08%	-9.96%	-1.33%	-9.04%	-14.78%	-1.79%	17.26%	15.71%	-0.30%	2.57%	-17.23%	-9.96%	-11.64%	7.67%	10.88%	-3.26%	-18.60%	6.25%	3.09%	3.33%	-15.43%	-14.40%	1.84%	5.83%	3.70%	1.70%	0.00%		
18.09%	14.31%	-8.95%	-9.94%	1.13%	11.30%	-5.48%	-22.22%	7.01%	-3.74%	19.05%	1.17%	6.47%	4.32%	28.21%	-4.44%	45.02%	30.90%	-37.70%	-1.48%	-20.14%	-1.50%	6.45%	28.66%	28.74%	6.18%	4.14%	-13.34%	-5.04%	44.85%	-6.47%	-0.00%	-26.78%	58.58%	9.59%	-11.48%	17.18%	-4.71%	7.27%	-14.78%	-7.95%	-9.80%	4.24%	11.12%	3.15%	0.00%		
12.98%	35.53%	31.87%	6.00%	13.54%	18.47%	-1.51%	6.20%	8.14%	20.40%	-1.02%	20.09%	-6.09%	-25.00%	25.46%	-6.35%	23.49%	3.51%	18.37%	0.82%	-6.69%	14.01%	25.23%	9.37%	21.52%	-0.61%	7.95%	5.29%	11.73%	-3.91%	55.19%	1.36%	14.22%	-8.26%	4.99%	11.53%	19.38%	-1.85%	53.67%	-13.50%	38.00%	4.75%	26.92%	-32.76%	8.26%	0.06%		
9.80%	2.37%	-1.49%	21.58%	-0.45%	4.69%	4.25%	21.11%	-4.23%	18.26%	13.51%	7.58%	0.57%	9.88%	-8.01%	-2.89%	6.01%	-10.88%	10.00%	-7.95%	7.40%	26.53%	12.26%	14.74%	7.43%	17.53%	13.05%	-6.09%	8.95%	-1.62%	-6.25%	5.19%	10.63%	4.70%	6.70%	6.40%	10.47%	11.79%	-6.03%	-11.99%	3.60%	22.49%	3.49%	4.80%	0.97%	0.00%		
11.87%	1.50%	7.88%	1.52%	5.14%	2.48%	31.68%	12.75%	10.79%	13.70%	9.99%	-15.28%	1.97%	14.67%	21.50%	1.10%	-17.32%	-6.00%	-2.01%	-7.97%	6.84%	7.14%	31.06%	16.10%	11.66%	30.16%	0.39%	-20.45%	-14.40%	-8.60%	-14.14%	-18.94%	3.08%	26.46%	-10.01%	16.47%	4.82%	-12.95%	33.64%	24.34%	11.66%	15.64%	2.68%	11.91%	0.13%			
13.85%	23.32%	20.81%	24.14%	21.02%	0.66%	19.11%	24.62%	-0.98%	12.69%	7.39%	23.18%	-5.55%	19.88%	11.26%	5.80%	-1.04%	24.03%	5.30%	-11.48%	37.81%	1.89%	19.63%	7.96%	23.16%	15.88%	-2.09%	-2.23%	6.90%	-1.34%	-1.19%	-0.08%	18.73%	21.04%	16.05%	13.57%	4.61%	32.41%	-8.19%	-8.18%	24.55%	2.99%	35.71%	26.79%	11.41%	0.21%		
25.43%	-37.00%	36.43%	-35.74%	34.73%	17.69%	37.09%	17.80%	31.47%	34.79%	54.79%	27.62%	32.73%	37.70%	34.14%	16.10%	36.97%	32.09%	16.40%	29.36%	47.40%	4.72%	10.63%	9.94%	55.66%	18.18%	16.62%	9.33%	-6.50%	7.51%	-1.26%	4.68%	58.48%	18.08%	21.64%	68.14%	41.39%	30.42%	-3.60%	8.23%	48.08%	24.50%	7.88%	18.86%	21.14%	0.21%		
16.05%	-10.73%	19.41%	25.44%	8.96%	15.75%	16.79%	37.73%	6.44%	36.91%	0.10%	-11.10%	-2.01%	25.20%	4.30%	21.79%	15.73%	32.08%	20.49%	32.90%	10.82%	19.40%	13.77%	7.80%	24.54%	-4.96%	15.73%	26.35%	1.17%	9.15%	7.84%	-32.97%	11.50%	41.05%	-7.91%	17.20%	28.36%	-4.57%	3.71%	-4.56%	3.72%	1.67%	8.86%	-2.82%	10.97%	0.12%		
13.61%	30.72%	6.30%	16.80%	14.67%	22.24%	9.52%	17.77%	14.64%	26.81%	28.00%	8.62%	0.80%	-7.85%	17.79%	7.41%	-15.81%	-4.83%	20.86%	2.12%	11.12%	1.88%	23.27%	19.27%	7.74%	38.23%	1.97%	1.32%	21.71%	10.78%	22.54%	24.49%	13.36%	4.19%	-16.97%	2.82%	13.57%	10.60%	3.05%	13.56%	6.10%	22.56%	18.07%	17.14%	10.05%	0.08%		
21.26%	-6.70%	-21.20%	23.90%	41.00%	41.52%	0.28%	-1.42%	61.97%	-47.78%	17.10%	-5.20%	-14.42%	-17.75%	58.80%	16.47%	7.96%	38.09%	96.13%	42.59%	6.46%	-50.18%	-57.49%	-14.78%	17.89%	-9.78%	61.37%	-1.71%	48.67%	16.44%	22.86%	13.75%	34.16%	7.52%	-0.68%	37.27%	15.63%	9.82%	10.34%	-38.77%	-26.21%	-21.37%	70.60%	11.99%	0.00%			
19.66%	31.64%	29.84%	8.29%	6.12%	-11.52%	-10.36%	38.46%	-14.20%	37.11%	12.46%	10.23%	0.80%	37.09%	14.25%	2.10%	17.69%	-36.59%	3.31%	-6.50%	30.25%	5.46%	-11.61%	-4.69%	11.95%	0.57%	12.84%	23.64%	49.72%	7.49%	-4.69%	-21.17%	13.08%	18.97%	26.88%	29.62%	-2.51%	53.17%	54.99%	-14.95%	7.48%	26.70%	10.00%	0.00%				
16.15%	-9.50%	24.26%	-1.51%	40.85%	41.41%	28.96%	7.27%	9.05%	23.49%	11.59%	-1.34%	10.96%	3.16%	-15.01%	7.34%	14.10%	18.11%	35.16%	6.44%	8.53%	-15.00%	18.01%	1.16%	18.08%	-15.76%	47.42%	34.84%	14.03%	2.86%	4.10%	8.01%	27.18%	39.61%	29.57%	33.94%	13.68%	28.77%	5.54%	13.30%	24.69%	26.79%	14.02%	0.11%				
17.54%	-1.62%	31.45%	16.59%	19.49%	14.78%	11.85%	7.68%	30.16%	-25.52%	18.61%	-2.14%	-10.99%	-7.52%	22.52%	6.36%	36.99%	14.64%	-1.19%	-29.62%	-6.03%	1.40%	-10.74%	6.96%	10.16%	9.36%	3.64%	-17.77%	8.58%	0.21%	14.78%	-7.88%	14.95%	13.32%	21.16%	19.86%	7.61%	7.91%	0.06%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%		
11.02%	0.00%	6.00%	4.00%	5.67%	3.75%	5.81%	7.59%	25.87%	9.53%	9.06%	13.96%	-0.28%	11.91%	-1.67%	26.19%	-6.04%	15.22%	-5.19%	-1.32%	-0.02%	9.85%	9.95%	1.27%	10.82%	9.98%	1.84%	4.66%	4.21%	20.43%	-1.30%	6.19%	1.18%	14.78%	18.00%	-16.00%	-12.22%	1.23%	3.29%	19.13%	5.68%	12.02%	8.37%	17.12%	-5.48%	9.80%	0.00%	
24.99%	3.97%	36.93%	37.69%	17.34%	3.74%	8.75%	15.21%	-1.89%	6.09%	7.55%	-4.75%	11.34%	9.12%	-1.03%	26.38%	2.14%	-26.00%	4.22%	3.4%	-41.22%	17.39%	26.92%	31.48%	34.66%	38.66%	-8.81%	-0.29%	14.94%	-6.00%	-0.23%	4.03%	6.84%	-4.51%	35.54%	-14.77%	27.83%	-4.43%	3.75%	-4.64%	-5.76%	30.20%	7.23%	2.41%	6.84%	0.00%		
11.61%	15.42%	5.17%	-0.68%	16.51%	-12.67%	18.56%	15.06%	5.29%	6.63%	15.55%	-1.78%	37.17%	-8.00%	19.59%	1.53%	16.73%	-6.01%	-1.62%	-4.56%	21.30%	11.87%	10.98%	22.88%	15.38%	1.98%	6.19%	3.34%	-33.29%	23.33%	18.61%	18.39%	-4.93%	-1.28%	-4.49%	15.01%	-1.28%	8.21%	14.55%	7.69%	23.64%	-6.04%	10.49%	27.41%	9.84%	0.04%		
5.90%	-5.28%	2.01%	6.42%	14.99%	-1.37%	7.85%	0.53%	0.21%	2.81%	5.97%	-0.15%	1.35%	-1.95%	-0.47%	2.44%	-4.06%	-1.80%	-1.15%	13.14%	-10.34%	4.10%	7.40%	1.29%	5.86%	-0.22%	-0.38%	4.75%	4.62%	0.60%	-0.17%	-0.76%	0.41%	-10.07%	3.49%	2.37%	1.56%	1.11%	1.74%	2.85%	-2.10%	6.90%	3.80%	-5.55%	1.38%	0.01%		
5.90%	-1.70%	6.60%	1.90%	-7.62%	-6.72%	6.11%	6.12%	5.26%	-1.38%	7.80%	5.60%	7.68%	7.78%	1.88%	4.49%	-0.68%	16.98%	-2.17%	7.11%	4.87%	13.40%	2.67%	2.68%	-4.35%	1.10%	-4.91%	11.43%	2.47%	2.20%	6.14%	10.47%	0.88%	0.78%	-0.27%	2.64%	7.62%	-2.89%	-1.72%	-0.22%	2.33%	3.38%	10.40%	-0.94%	1.17%	0.00%		
5.90%	3.76%	-11.03%	-4.58%	13.64%	-1.25%	13.37%	12.46%	18.78%	1.74%	6.79%	-2.49%	-1.24%	-4.02%	-4.36%	-0.74%	-4.13%	1.02%	-1.38%	-1.44%	10.41%	-2.39%	-4.03%	5.95%	8.36%	12.07%	0.71%	10.52%	-7.20%	6.17%	1.61%	6.81%	3.26%	-4.05%	6.46%	8.29%	12.91%	6.56%	5.78%	-0.54%	-4.88%	4.96%	4.91%	1.96%	1.84%	0.00%		
5.90%	4.05%	-0.34%	-1.08%	13.68%	17.02%	-1.81%	-4.02%	4.38%	-10.80%	3.24%	-2.95%	-2.45%	12.79%	-0.08%	1.38%	-0.81%	12.23%	3.44%	-3.20%	10.89%	-1.95%	-8.30%	1.38%	-3.39%	19.02%	-5.48%	1.98%	-5.48%	1.98%	-5.48%	1.98%	-5.48%	1.98%	-5.48%	1.98%	-5.48%	1.98%	-5.48%	1.98%	-5.48%	1.98%	-5.48%	1.98%	-5.48%	1.98%	1.84%	0.00%
5.90%	-1.79%	-0.48%	6.29%	3.10%	4.68%	1.01%	2.25%	-1.27%	1.74%	6.86%	-0.63%	-1.29%	2.59%	2.65%	-0.29%	-1.81%	-1.05%	0.50%	14.05%	-10.34%	-1.36%	3.73%	3.24%	1.86%	-2.15%	4.71%	3.00%	8.20%	1.71%	4.12%	3.88%	3.24%	8.46%	14.70%	6.45%	4.78%	0.42%	-1.70%	7.95%	1.06%	9.58%	-4.88%	-				

References for Portfolio Returns Calculated		
Energy XLE		https://finance.yahoo.com/quote/XLE/performance/
Agriculture(DBA)	Invesco DB Agriculture Fund (DBA)	https://finance.yahoo.com/quote/DBA/performance/
Broad Commodity(DJP)	iPath Bloomberg Commodity Index Total Return(SM) ETN (DJP)	https://finance.yahoo.com/quote/DJP/performance/
Canadian Balanced Fund	iShares S&P/TSX 60 Index ETF (XIU.TO)	https://finance.yahoo.com/quote/XIU.TO/performance?p=XIU.TO
Global Balanced Fund	JPM Global Balanced C (acc) USDH (0P000115RQ)	https://finance.yahoo.com/quote/0P000115RQ/summary/
Canadian Equity Fund	Canada Life Canadian Dividend A (0P0001MKV3.TO)	https://ca.finance.yahoo.com/quote/0P0001MKV3.TO/performance?p=0P0001MKV3.TO
US Equity Fund	Schwab U.S. Dividend Equity ETF (SCHD)	https://finance.yahoo.com/quote/SCHD/performance?p=SCHD
Technology funds	SPDR S&P Software & Services ETF (XSW)	https://finance.yahoo.com/quote/XSW/performance?p=XSW
Financial Services(BFSI)	iShares Global Financials ETF (IXG)	https://finance.yahoo.com/quote/IXG/performance?p=IXG
Healthcare	iShares U.S. Healthcare Providers ETF (IHF)	https://finance.yahoo.com/quote/IHF/performance?p=IHF
Energy sectoral fund	Vanguard Energy Index Fund (VDE)	https://finance.yahoo.com/quote/VDE/performance?p=VDE
Consumer discretionary	iShares US Consumer Discretionary ETF (IYC)	https://finance.yahoo.com/quote/IYC/performance?p=IYC
Industrial sector	Invesco S&P 500 Equal Weight ETF (RSP)	https://finance.yahoo.com/quote/RSP/performance?p=RSP
Materials Sector	Vanguard Materials Index Fund (VAW)	https://finance.yahoo.com/quote/VAW/performance?p=VAW
Consumer Staples	iShares Global Consumer Staples ETF (KXI)	https://finance.yahoo.com/quote/KXI/performance?p=KXI
Telecommunications	SPDR S&P Telecom ETF (XTL)	https://finance.yahoo.com/quote/XTL/performance?p=XTL
Utility Funds	Franklin Utilities A1 (FKUTX)	https://finance.yahoo.com/quote/FKUTX/performance?p=FKUTX
International soverirgn bonds	iShares International Treasury Bond ETF (IGOV)	https://finance.yahoo.com/quote/IGOV/performance?p=IGOV
Provincial/Municipal Bonds, 10%	Horizons Active Cdn Municipal Bond ETF (HMP.TO)	https://ca.finance.yahoo.com/quote/HMP.TO/performance?p=HMP.TO
Investment-Grade Corporate Bonds, 40%	MIL BlackRock Global Corporate Bond Fund	https://finance.yahoo.com/quote/0P00012AQT/performance?p=0P00012AQT
High-Yield (Junk) Bonds, 20%	American Funds American High-Income Trust (AHTFX)	https://finance.yahoo.com/quote/AHTFX/performance?p=AHTFX
Emerging Market Corporate Bonds, 20%	Vanguard Emerging Markets Government Bond Index Fund (VWOB)	https://finance.yahoo.com/quote/VWOB/performance?p=VWOB
Convertible Bonds, 10%	Morgan Stanley Investment Funds - Global Convertible Bond Fund I (0P00000SJ3)	https://finance.yahoo.com/quote/0P00000SJ3/performance?p=0P00000SJ3
Preferred Shares, 10%	Cohen & Steers Low Duration Pref & Inc	https://finance.yahoo.com/quote/LPXIX/performance?p=LPXIX
Leveraged Loans, 30%	Direxion Daily 20+ Year Treasury Bull 3X Shares (TMF)	https://finance.yahoo.com/quote/TMF/performance?p=TMF
Collateralized Loan Obligations (CLOs)	AXS First Priority CLO Bond ETF (AAA)	https://finance.yahoo.com/quote/AAA/performance?p=AAA
Mortgage-Backed Securities (MBS)	iShares MBS ETF (MBB)	https://finance.yahoo.com/quote/MBB/performance?p=MBB
Treasury Bills	iShares Treasury Floating Rate Bond ETF (TFLO)	https://finance.yahoo.com/quote/TFLO/performance?p=TFLO
Blue Bonds	Oersted Blue-Bond Issue	https://orsted.com/en/media/newsroom/news/2023/06/20230608684811
Segregated Bonds	Canada Life Balanced Allocation	https://www.canadalife.com/content/dam/canadalife/documents/investing-savings/segregated-funds/canada-life/rates-of-return/en/cl-estate.pdf
Private Equity - Central Returns	BlackRock	https://www.blackrock.com/institutions/en-us/insights/charts/capital-market-assumptions
Infra Projects	Global 60/40 Allocation	https://www.blackrock.com/institutions/en-us/insights/charts/capital-market-assumptions#assumptions
Social Bonds	GDP Growth Assumptions - JP Morgan Assumptions	https://am.jpmorgan.com/content/dam/jpm-am-aem/global/en/insights/portfolio-insights/ltcma/noindex/ltcma-full-report.pdf

Portfolio Returns – Value at Risk: Historical Approach

VALUE AT RISK MODEL - PENSION FUND																													
Date	Adj Close	Returns	Sorted Returns	Calculation of Value at Risk - Fund(Historical)																									
2/3/2020	11707.90039	2.49%	2.49%	<table><tr><td>Mean</td><td>5.44%</td></tr><tr><td>Std Deviation</td><td>1.87%</td></tr><tr><td>Min</td><td>1.42%</td></tr><tr><td>Max</td><td>10.34%</td></tr><tr><td>CMP</td><td>1000000.00</td></tr></table>	Mean	5.44%	Std Deviation	1.87%	Min	1.42%	Max	10.34%	CMP	1000000.00															
Mean	5.44%																												
Std Deviation	1.87%																												
Min	1.42%																												
Max	10.34%																												
CMP	1000000.00																												
2/20/2020	12080.84961	5.45%	5.45%																										
2/13/2020	12174.65039	5.56%	5.56%																										
2/5/2020	12089.15039	5.38%	5.38%																										
3/5/2020	11269	6.63%	6.63%																										
3/4/2020	11251	1.76%	1.76%	<table><tr><th>Percentile</th><th>Confidence</th><th>VAR(%)</th><th>Fund Valu</th><th>VAR(INR)</th></tr><tr><td>5%</td><td>95%</td><td>2.53%</td><td>974680.00</td><td>25320.00</td></tr><tr><td>1%</td><td>99%</td><td>1.56%</td><td>984372.00</td><td>15628.00</td></tr><tr><td>0.50%</td><td>99.50%</td><td>1.49%</td><td>985086.00</td><td>14914.00</td></tr><tr><td>10%</td><td>90.00%</td><td>3.23%</td><td>967660.00</td><td>32340.00</td></tr></table>	Percentile	Confidence	VAR(%)	Fund Valu	VAR(INR)	5%	95%	2.53%	974680.00	25320.00	1%	99%	1.56%	984372.00	15628.00	0.50%	99.50%	1.49%	985086.00	14914.00	10%	90.00%	3.23%	967660.00	32340.00
Percentile	Confidence	VAR(%)	Fund Valu		VAR(INR)																								
5%	95%	2.53%	974680.00		25320.00																								
1%	99%	1.56%	984372.00		15628.00																								
0.50%	99.50%	1.49%	985086.00		14914.00																								
10%	90.00%	3.23%	967660.00	32340.00																									
2/27/2020	11633.29981	10.34%	10.34%																										
2/25/2020	11797.90039	3.57%	3.57%																										
3/6/2020	10989.4502	6.49%	6.49%																										
3/3/2020	11303.29981	5.29%	5.29%																										
2/19/2020	12125.90039	4.94%	4.94%																										
2/11/2020	12107.90039	4.08%	4.08%																										
2/6/2020	12137.9502	7.47%	7.47%																										
2/18/2020	11992.5	7.61%	7.61%																										
2/12/2020	12201.2002	5.87%	5.87%																										
2/10/2020	12031.5	3.98%	3.98%																										
2/4/2020	11979.65039	5.81%	5.81%																										
3/12/2020	9590.150391	7.77%	7.77%																										
6/23/2020	10471	4.70%	4.70%																										
6/29/2020	10312.40039	6.57%	6.57%																										
6/24/2020	10305.29981	3.23%	3.23%																										
6/11/2020	9902	7.63%	7.63%																										
6/17/2020	9881.150391	3.49%	3.49%																										
4/30/2020	9859.900391	4.83%	4.83%																										
6/4/2020	10029.09961	1.42%	1.42%																										
6/9/2020	10046.65039	6.56%	6.56%																										
6/25/2020	10288.90039	3.66%	3.66%																										
3/2/2020	11132.75	2.91%	2.91%																										
5/29/2020	9580.299805	8.33%	8.33%																										
3/19/2020	8263.450195	4.96%	4.96%																										
4/8/2020	8748.75	5.35%	5.35%																										
6/15/2020	9813.700195	6.64%	6.64%																										
3/11/2020	10458.40039	4.59%	4.59%																										
6/22/2020	10311.2002	8.06%	8.06%																										
6/1/2020	9826.150391	5.84%	5.84%																										
5/7/2020	9199.049805	6.58%	6.58%																										
5/12/2020	9196.549805	3.25%	3.25%																										
2/17/2020	12045.79981	4.32%	4.32%																										
5/6/2020	9270.900391	6.50%	6.50%																										
5/8/2020	9251.5	5.11%	5.11%																										
5/13/2020	9383.549805	4.96%	4.96%																										
6/19/2020	10244.40039	5.95%	5.95%																										
5/27/2020	9314.950195	7.97%	7.97%																										

Portfolio Returns – Value at Risk: Montecarlo Approach

Returns	Sorted Returns	Replication	Simulated Returns	Calculation of Value at Risk - Pension Fund			
2.49%	2.49%	1	0.027390131				
5.45%	5.45%	2	0.085315391	Historical Approach			
5.56%	5.56%	3	0.033401571	Mean		5.44%	
5.38%	5.38%	4	0.044182731	Std Deviation		1.87%	
6.63%	6.63%	5	0.053376606	Min		1.42%	
1.76%	1.76%	6	0.045182113	Max		10.34%	
10.34%	10.34%	7	0.04325508	CMP		19794.00	
3.57%	3.57%	8	0.046374937				
6.49%	6.49%	9	0.035337694	Percentile	Confidence	VAR(%)	Stock Price VAR(INR)
5.29%	5.29%	10	0.056325011	5%	95%	2.53%	19292.82 501.18
4.94%	4.94%	11	0.067032707	1%	99%	1.56%	19484.66 309.34
4.08%	4.08%	12	0.047484313	0.50%	99.50%	1.49%	19498.79 295.21
7.47%	7.47%	13	0.073152069	10%	90.00%	3.23%	19153.86 640.14
7.61%	7.61%	14	0.094823635				
5.87%	5.87%	15	0.046972372	MonteCarlo Simulation			
3.98%	3.98%	16	0.085238445	Mean		5.59%	
5.81%	5.81%	17	0.023963215	Std Deviation		1.91%	
7.77%	7.77%	18	0.06589667	Min		2.40%	
4.70%	4.70%	19	0.060074848	Max		8.33%	
6.57%	6.57%	20	0.064148687	CMP		#####	
3.23%	3.23%	21	0.059471024				
7.63%	7.63%	22	0.056453958	Percentile	Confidence	VAR(%)	Stock Price VAR(INR)
3.49%	3.49%	23	0.061369378	5%	95%	2.75%	19250.18 543.82
4.83%	4.83%	24	0.081775365	1%	99%	2.44%	19310.73 483.27
1.42%	1.42%	25	0.025065168	0.50%	99.50%	2.42%	19315.31 478.69
6.56%	6.56%	26	0.061888602	10%	90.00%	3.48%	19105.34 688.66
3.66%	3.66%	27	0.065239771				
2.91%	2.91%	28	0.042712453				
8.33%	8.33%	29	0.028230649				
4.96%	4.96%	30	0.061217292				
5.35%	5.35%	31	0.029873448				
6.64%	6.64%	32	0.077208492				
4.59%	4.59%	33	0.104625914				
8.06%	8.06%	34	0.071486573				
5.84%	5.84%	35	0.046372697				
6.58%	6.58%	36	0.04161522				
3.25%	3.25%	37	0.041169485				
4.32%	4.32%	38	0.058226189				
6.50%	6.50%	39	0.061374099				
5.11%	5.11%	40	0.048810317				
4.96%	4.96%	41	0.036218435				
5.95%	5.95%	42	0.075707191				
7.97%	7.97%	43	0.06771439				

Muskrat Falls – Statement of Financial Position

<i>As at (thousands of Canadian dollars)</i>	<i>Notes</i>	March 31 2022	December 31 2021
ASSETS			
Current assets			
Restricted cash		989,036	420,674
Trade and other receivables		8,160	3,993
Related party loan receivable	12	2,403	9,843
Short-term investments	4	33,872	-
Contract asset	12	31,525	17,573
Other current assets		1,113	2,600
Total current assets		1,066,109	454,683
Non-current assets			
Property, plant and equipment	3	6,415,105	6,398,742
Investments	4	234,928	-
Other long-term asset	12	5,189	3,014
Total assets		7,721,331	6,856,439
LIABILITIES AND EQUITY			
Current liabilities			
Trade and other payables	5	203,815	169,609
Current portion of long-term debt	6	33,872	33,872
Current portion of long-term payables	7	40,468	44,309
Related party loan payable	12	2,404	53,220
Total current liabilities		280,559	301,010
Non-current liabilities			
Long-term debt	6	4,464,460	3,630,194
Long-term payables	7	108,813	88,795
Total liabilities		4,853,832	4,019,999
Shareholder's equity			
Share capital	8	1,847,725	1
Shareholder contributions		1,166,594	3,011,317
Reserves		(47,385)	(48,143)
Deficit		(99,435)	(126,735)
Total equity		2,867,499	2,836,440
Total liabilities and equity		7,721,331	6,856,439

Muskrat Falls – Statement of P&L and Comprehensive Income

<i>For the period ended March 31 (thousands of Canadian dollars)</i>	Notes	Three months ended	
		2022	2021
Revenue		121,038	450
Total revenue		121,038	450
Power purchased	12	3,001	-
Operating costs	9	7,381	1,367
Transmission expense		33,393	38,726
Depreciation		56	-
Net finance expense	10	29,222	-
Other expense		3,703	1,153
Total expenses		76,756	41,246
Profit (loss) for the period		44,282	(40,796)
Other comprehensive income			
Total items that may or have been reclassified to profit or loss:			
Reclassification adjustment related to:			
Cash flow hedges recognized in profit or loss		758	764
Other comprehensive income for the period		758	764
Total comprehensive income (loss) for the period		45,040	(40,032)

Muskrat Falls – Statement of Changes in Equity

<i>(thousands of Canadian dollars)</i>	Notes	Share Capital	Shareholder Contributions	Reserves	Deficit	Total
Balance at January 1, 2022		1	3,011,317	(48,143)	(126,735)	2,836,440
Profit for the period		-	-	-	44,282	44,282
Other comprehensive income		-	-	758	-	758
Total comprehensive income for the period		-	-	758	44,282	45,040
Shareholder contributions	12	-	3,001	-	-	3,001
Capital restructuring	8	1,847,724	(1,847,724)	-	-	-
Related party transaction	12	-	-	-	(16,982)	(16,982)
Balance at March 31, 2022		1,847,725	1,166,594	(47,385)	(99,435)	2,867,499
Balance at January 1, 2021		1	2,783,975	(51,189)	(14,852)	2,717,935
Loss for the period		-	-	-	(40,796)	(40,796)
Other comprehensive income		-	-	764	-	764
Total comprehensive loss for the period		-	-	764	(40,796)	(40,032)
Balance at March 31, 2021		1	2,783,975	(50,425)	(55,648)	2,677,903

Muskrat Falls – Statement of Cash Flows

<i>For the period ended March 31 (thousands of Canadian dollars)</i>	Notes	Three months ended	
		2022	2021
Operating activities			
Profit (loss) for the period		44,282	(40,796)
Adjustments to reconcile profit (loss) to cash provided from (used in) operating activities:			
Depreciation and amortization		814	-
Revenue	12	(16,982)	-
Power purchased	12	3,001	-
Finance income	10	(699)	(553)
Finance expense	10	29,921	553
		60,337	(40,796)
Changes in non-cash working capital balances	14	(3,934)	1,016
Increase in long-term payables		16,177	38,726
Increase in contract asset		(13,952)	-
Increase in other long-term asset		(2,175)	-
Interest received		622	535
Interest paid		(656)	(2)
Net cash provided from (used in) operating activities		56,419	(521)
Investing activities			
Increase in investments	4	(268,800)	-
Decrease in advances		820	10
Additions to property, plant and equipment		(20,007)	(49,597)
Changes in non-cash working capital balances	14	8,781	(12,928)
Net cash used in investing activities		(279,206)	(62,515)
Financing activities			
Change in restricted cash		(568,362)	63,036
Repayment of related party loan payable		(50,539)	-
Decrease in related party loan receivable		7,400	-
Proceeds from long-term debt		834,288	-
Net cash provided from financing activities		222,787	63,036
Net increase (decrease) in cash		-	-
Cash, beginning of the period		-	-
Cash, end of the period		-	-